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LIPIDS, FORMULATIONS, AND USES THEREOF

Abstract

The present disclosure provides compounds, compositions, formulations, and methods for delivery of active agents (e.g., nucleic acid). In particular, the present disclosure provides modified lipids (e.g., lipids conjugated with 1-methyl-D-tryptophan) for use in delivery of active agents and compositions and vaccines thereof.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/334,437, filed Apr. 25, 2022, the content of which is herein incorporated by reference in its entirety.

FIELD

[0002] The present disclosure provides compounds, compositions, formulations, and methods for delivery of active agents (e.g., nucleic acid).

BACKGROUND

[0003] It is of great interest for therapeutics, diagnostics, reagents and for biological assays to be able to deliver a macromolecules and active agents (e.g., a nucleic acid), into a cell. Recently, delivery of siRNA and mRNA based therapeutics are now in clinical use against coronavirus disease 2019 (COVID-19) and cancer. However, delivery of nucleic acids is difficult due to instability, low permeability, and unwanted activation of the innate immune response. Safe, effective, and stable delivery systems are needed to facilitate the in vivo delivery of nucleic acids.

[0004] Lipid nanoparticles (LNP) and cationic liposomes have already been proven as effective delivery systems for nucleic acids. The composition of lipid nanoparticle or cationic liposomes includes cationic lipids or ionizable lipids and other types of lipids, in which cationic or ionizable lipids are functionalized to encapsulate nucleic acid and induce efficient transfection in vivo.

Though a variety of such lipid nanoparticle compositions have been demonstrated, improvements in safety, efficacy, and specificity are still lacking.

SUMMARY

[0005] In one aspect, disclosed herein is a compound of formula (I):

##STR00001## [0006] or a pharmaceutically acceptable salt thereof, wherein: [0007] R^{sup.1} is hydrogen, or C_{sub.1}-C_{sub.6} alkyl; [0008] X is selected from O, NR^{sup.2}, S, and a bond; [0009] R^{sup.2} is hydrogen C_{sub.1}-C_{sub.6} alkyl, C_{sub.2}-C_{sub.6} alkenyl, C_{sub.2}-C_{sub.6} alkynyl, C_{sub.3}-C_{sub.6} cycloalkyl, or halo-C_{sub.1}-C_{sub.6}-alkyl; and [0010] L is selected from C_{sub.8}-C_{sub.80} alkyl, C_{sub.8}-C_{sub.80} alkenyl, C_{sub.8}-C_{sub.80} alkynyl, C_{sub.8}-C_{sub.80} heteroalkyl, C_{sub.8}-C_{sub.80} heteroalkenyl, and C_{sub.8}-C_{sub.80} heteroalkynyl, each of which is optionally substituted with one or more substituents selected from hydroxy and amino.

[0011] In some embodiments, R^{sup.1} is methyl

[0012] In some embodiments, L is selected from C_{sub.12}-C_{sub.40} alkyl and C_{sub.12}-C_{sub.40} alkenyl. In some embodiments, L has a formula (A):

##STR00002## [0013] wherein: [0014] n is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, or 40; and [0015] R^{sup.a} and R^{sup.b} are each independently selected from C_{sub.6}-C_{sub.40} alkyl, C_{sub.6}-C_{sub.40} alkenyl, C_{sub.6}-C_{sub.40} heteroalkyl, and C_{sub.6}-C_{sub.40} heteroalkenyl.

In some embodiments, n is 1, 2, 3, 4, 5, 6, 7, or 8.

[0016] In some embodiments, L has a formula (D), (E), or (F):

##STR00003## [0017] wherein: [0018] n, p, and q are each independently 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, or 40; and [0019] R^{sup.a1} and R^{sup.a2} are each independently selected from C_{sub.6}-C_{sub.40} alkyl and C_{sub.6}-C_{sub.40} alkenyl.

In some embodiments, n, p, and q are each independently 1, 2, 3, 4, 5, 6, 7, or 8.

[0020] In some embodiments, L is selected from:

##STR00004## ##STR00005## ##STR00006##

[0021] In some embodiments, the compound is selected from the group consisting of:

##STR00007##

[0022] In another aspect, disclosed is a lipid composition comprising a compound disclosed herein (e.g., a compound of formula (1)) or a pharmaceutically acceptable salt thereof. In some embodiments, the lipid composition further comprises one or more of a cationic and/or ionizable lipid, a phospholipid, a neutral or non-cationic lipid, or a combination thereof.

[0023] In some embodiments, the lipid composition further comprises a structural lipid. In some embodiments, the structural lipid is selected from the group consisting of cholesterol, fecosterol, sitosterol, ergosterol, campesterol, stigmasterol, brassicasterol, tomatidine, cholesteryl hemisuccinate, cholesteryl sulfate, ursolic acid, alpha-tocopherol, and mixtures thereof.

[0024] In some embodiments, the lipid composition further comprises a polyethylene glycol (PEG)-lipid conjugate.

[0025] In some embodiments, the lipid composition further comprises at least one active agent. In some embodiments, the at least one active agent is encapsulated by a liposome, a lipid nanoparticle, a micelle in the lipid composition. In some embodiments, the at least one active agent comprises a nucleic acid or protein. In some embodiments, the nucleic acid encodes an antigen or a functional fragment thereof.

[0026] In some embodiments, the at least one active agent comprises an RNA. In some embodiments, the RNA is selected from the group consisting of a small interfering RNA (siRNA), an asymmetrical interfering RNA (aiRNA), a microRNA (miRNA), a Dicer-substrate RNA (dsRNA), a small hairpin RNA (shRNA), a messenger RNA (mRNA), a ribozyme, and mixtures thereof.

[0027] In some embodiments, the at least one active agent comprises an immune modulator, a chemotherapeutic agent, a steroid, an analgesic, an antimicrobial agent, or a combination thereof.

[0028] In another aspect, disclosed herein is a pharmaceutical composition comprising an effective amount of a compound disclosed herein (e.g., a compound of formula (I)) or a pharmaceutically acceptable salt thereof, or a lipid composition disclosed herein, and a pharmaceutically acceptable carrier.

[0029] In another aspect, disclosed herein is a vaccine comprising an effective amount of: a compound disclosed herein (e.g., a compound of formula (I), or a pharmaceutically acceptable salt thereof), a lipid composition as described herein, or a pharmaceutical composition disclosed herein (e.g., a pharmaceutical composition comprising a compound of formula (I), or a pharmaceutically acceptable salt thereof, or a lipid composition as described herein); and an antigen or a nucleic acid encoding thereof.

[0030] In some embodiments, the antigen is a tumor antigen, a self-antigen, or an infectious disease derived antigen. In some embodiments, the nucleic acid is messenger RNA (mRNA).

[0031] In another aspect, disclosed herein is a method for delivering an active agent to a cell comprising contacting the cell a lipid composition disclosed herein, or a pharmaceutical composition disclosed herein (e.g., a pharmaceutical composition comprising a compound of formula (I), or a pharmaceutically acceptable salt thereof, or a lipid composition as described herein), or a vaccine disclosed herein (e.g., a vaccine comprising a compound of formula (I), or a pharmaceutically acceptable salt thereof, or a lipid composition as described herein), to a subject in need thereof.

[0032] In another aspect, disclosed herein is a method for delivering an active to a subject comprising administering to the subject a lipid composition disclosed herein, or a pharmaceutical composition disclosed herein (e.g., a pharmaceutical composition comprising a compound of formula (I), or a pharmaceutically acceptable salt thereof, or a lipid composition as described herein), or a vaccine disclosed herein (e.g., a vaccine comprising a compound of formula (I), or a pharmaceutically acceptable salt thereof, or a lipid composition as described herein), to a subject in

need thereof.

[0033] In another aspect, disclosed herein is a method for producing a polypeptide of interest in a cell comprising contacting the cell a lipid composition disclosed herein, or a pharmaceutical composition disclosed herein (e.g., a pharmaceutical composition comprising a compound of formula (I), or a pharmaceutically acceptable salt thereof, or a lipid composition as described herein), or a vaccine disclosed herein (e.g., a vaccine comprising a compound of formula (I), or a pharmaceutically acceptable salt thereof, or a lipid composition as described herein), to a subject in need thereof, wherein the lipid composition or pharmaceutical composition comprises an mRNA encoding the polypeptide of interest. In some embodiments, the cell is in a subject and the contacting comprises administering to the subject.

[0034] In another aspect, disclosed herein is a method of treating or preventing a disease or disorder comprising administering an effective amount of a compound disclosed herein (e.g., a compound of formula (I), or a pharmaceutically acceptable salt thereof), or a lipid composition disclosed herein, or a pharmaceutical composition disclosed herein (e.g., a pharmaceutical composition comprising a compound of formula (I), or a pharmaceutically acceptable salt thereof, or a lipid composition as described herein), or a vaccine disclosed herein (e.g., a vaccine comprising a compound of formula (I), or a pharmaceutically acceptable salt thereof, or a lipid composition as described herein), to a subject in need thereof.

[0035] In some embodiments, the disease or disorder comprises cancer, an autoimmune disease, an inflammatory disease, or an infectious disease. In some embodiments, the disease or disorder is cancer. In some embodiments, the subject has cancer, has had cancer, is predisposed to cancer, or has a family history of cancer. In some embodiments, the cancer comprises a solid tumor or hematological cancer. In some embodiments, the cancer is metastatic cancer. In some embodiments, the method suppresses or eliminates cancer metastasis, decreases tumor growth, prevents tumor recurrences, or any combination thereof. In some embodiments, the administering comprises an initial immunization and at least one subsequent immunization.

[0036] In some embodiments, the subject is human.

[0037] In some embodiments, the methods further comprise administering at least one additional active agent. In some embodiments, the at least one additional active agent comprises an immune modulator, a chemotherapeutic agent, a nucleic acid, a steroid, an analgesic, an antimicrobial agent, or a combination thereof.

[0038] Other aspects and embodiments of the disclosure will be apparent in light of the following detailed description.

Description

DETAILED DESCRIPTION

[0039] Described herein are modified lipids for use in liposomal and lipid nanoparticle delivery of active agents (e.g., miRNA) which facilitate efficient encapsulation, improved delivery, and increased downstream efficacy, e.g., due to macrophage polarization. The modified lipids comprise a cationic or ionizable lipid conjugated to an amino acid or an IDO inhibitor (e.g., 1-methyl-D-tryptophan).

[0040] Section headings as used in this section and the entire disclosure herein are merely for organizational purposes and are not intended to be limiting.

1. Definitions

[0041] The terms “comprise(s),” “include(s),” “having,” “has,” “can,” “contain(s),” and variants thereof, as used herein, are intended to be open-ended transitional phrases, terms, or words that do not preclude the possibility of additional acts or structures. The singular forms “a,” “and,” and “the” include plural references unless the context clearly dictates otherwise. The present disclosure

also contemplates other embodiments “comprising,” “consisting of,” and “consisting essentially of,” the embodiments or elements presented herein, whether explicitly set forth or not.

[0042] For the recitation of numeric ranges herein, each intervening number there between with the same degree of precision is explicitly contemplated. For example, for the range of 6-9, the numbers 7 and 8 are contemplated in addition to 6 and 9, and for the range 6.0-7.0, the number 6.0, 6.1, 6.2, 6.3, 6.4, 6.5, 6.6, 6.7, 6.8, 6.9, and 7.0 are explicitly contemplated.

[0043] Unless otherwise defined herein, scientific, and technical terms used in connection with the present disclosure shall have the meanings that are commonly understood by those of ordinary skill in the art. The meaning and scope of the terms should be clear; in the event, however of any latent ambiguity, definitions provided herein take precedent over any dictionary or extrinsic definition. Further, unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular. All publications, patent applications, patents and other references mentioned herein are incorporated by reference in their entirety.

[0044] The term “contacting” as used herein refers to bring or put in contact, to be in or come into contact. The term “contact” as used herein refers to a state or condition of touching or of immediate or local proximity.

[0045] “Polynucleotide,” “oligonucleotide,” or “nucleic acid,” as used herein, means at least two nucleotides covalently linked together. The polynucleotide may be DNA, both genomic and cDNA, RNA, or a hybrid, where the polynucleotide may contain combinations of deoxyribo- and ribonucleotides, and combinations of bases including uracil, adenine, thymine, cytosine, guanine, inosine, xanthine hypoxanthine, isocytosine and isoguanine. The nucleic acid, whether DNA or RNA may comprise non-natural nucleotides, modified nucleotides, and/or non-nucleotide building blocks that can exhibit the same function as natural nucleotides (e.g., “nucleotide analogs”). Nucleic acids may be obtained by chemical synthesis methods or by recombinant methods. Polynucleotides may be single- or double-stranded or may contain portions of both double stranded and single stranded sequence. The depiction of a single strand also defines the sequence of the complementary strand. Thus, a nucleic acid also encompasses the complementary strand of a depicted single strand. Many variants of a nucleic acid may be used for the same purpose as a given nucleic acid. Thus, a nucleic acid also encompasses substantially identical nucleic acids and complements thereof.

[0046] A “peptide” or “polypeptide” is a linked sequence of two or more amino acids linked by peptide bonds. Peptides and polypeptides include proteins such as binding proteins, receptors, and antibodies. The terms “polypeptide” and “protein” are used interchangeably herein.

[0047] As used herein, “nucleic acid” or “nucleic acid sequence” refers to a polymer or oligomer of pyrimidine and/or purine bases, preferably cytosine, thymine, and uracil, and adenine and guanine, respectively (See Albert L. Lehninger, Principles of Biochemistry, at 793-800 (Worth Pub. 1982)). The present technology contemplates any deoxyribonucleotide, ribonucleotide, or peptide nucleic acid component, and any chemical variants thereof, such as methylated, hydroxymethylated, or glycosylated forms of these bases, and the like. The polymers or oligomers may be heterogenous or homogenous in composition and may be isolated from naturally occurring sources or may be artificially or synthetically produced. In addition, the nucleic acids may be DNA or RNA, or a mixture thereof, and may exist permanently or transitionally in single-stranded or double-stranded form, including homoduplex, heteroduplex, and hybrid states. In some embodiments, a nucleic acid or nucleic acid sequence comprises other kinds of nucleic acid structures such as, for instance, a DNA/RNA helix, peptide nucleic acid (PNA), morpholino nucleic acid (see, e.g., Braasch and Corey, Biochemistry, 41 (14): 4503-4510 (2002)) and U.S. Pat. No. 5,034,506), locked nucleic acid (LNA; see Wablestedt et al., Proc. Natl. Acad. Sci. U.S.A., 97:5633-5638 (2000)), cyclohexenyl nucleic acids (see Wang, J. Am. Chem. Soc., 122:8595-8602 (2000)), and/or a ribozyme. Hence, the term “nucleic acid” or “nucleic acid sequence” may also encompass a chain comprising non-natural nucleotides, modified nucleotides, and/or non-nucleotide building blocks that can exhibit

the same function as natural nucleotides (e.g., “nucleotide analogs”); further, the term “nucleic acid sequence” as used herein refers to an oligonucleotide, nucleotide or polynucleotide, and fragments or portions thereof, and to DNA or RNA of genomic or synthetic origin, which may be single or double-stranded, and represent the sense or antisense strand. The terms “nucleic acid,” “polynucleotide,” “nucleotide sequence,” and “oligonucleotide” are used interchangeably. They refer to a polymeric form of nucleotides of any length, either deoxyribonucleotides or ribonucleotides, or analogs thereof.

[0048] As used herein, the terms “providing,” “administering,” and “introducing” are used interchangeably herein and refer to the placement of active agents into a subject by a method or route which results in at least partial localization to a desired site.

[0049] A “subject” or “patient” may be human or non-human and may include, for example, animal strains or species used as “model systems” for research purposes, such a mouse model as described herein. Likewise, patient may include either adults or juveniles (e.g., children). Moreover, patient may mean any living organism, preferably a mammal (e.g., humans and non-humans) that may benefit from the administration of compositions contemplated herein. Examples of mammals include, but are not limited to, any member of the Mammalian class: humans, non-human primates such as chimpanzees, and other apes and monkey species; farm animals such as cattle, horses, sheep, goats, swine; domestic animals such as rabbits, dogs, and cats; laboratory animals including rodents, such as rats, mice and guinea pigs, and the like. Examples of non-mammals include, but are not limited to, birds, fish, and the like. In one embodiment, the mammal is a human.

[0050] As used herein, “treat,” “treating,” and the like means a slowing, stopping, or reversing of progression of a disease or disorder when provided a compound or composition described herein to an appropriate control subject. The term also means a reversing of the progression of such a disease or disorder to a point of eliminating or greatly reducing the symptoms. As such, “treating” means an application or administration of the compositions described herein to a subject, where the subject has a disease or a symptom of a disease, where the purpose is to cure, heal, alleviate, relieve, alter, remedy, ameliorate, improve, or affect the disease or symptoms of the disease.

[0051] Definitions of specific functional groups and chemical terms are described in more detail below. For purposes of this disclosure, the chemical elements are identified in accordance with the Periodic Table of the Elements, CAS version, Handbook of Chemistry and Physics, 75th Ed., inside cover, and specific functional groups are generally defined as described therein.

Additionally, general principles of organic chemistry, as well as specific functional moieties and reactivity, are described in Sorrell, Organic Chemistry, 2nd edition, University Science Books, Sausalito, 2006; Smith, March's Advanced Organic Chemistry: Reactions, Mechanism, and Structure, 7th Edition, John Wiley & Sons, Inc., New York, 2013; Larock, Comprehensive Organic Transformations, 3rd Edition, John Wiley & Sons, Inc., New York, 2018; and Carruthers, Some Modern Methods of Organic Synthesis, 3rd Edition, Cambridge University Press, Cambridge, 1987; the entire contents of each of which are incorporated herein by reference.

[0052] The term “alkyl,” as used herein, means a straight or branched, saturated hydrocarbon chain. Representative examples of alkyl include, but are not limited to, methyl, ethyl, n-propyl, isopropyl, n-butyl, sec-butyl, iso-butyl, tert-butyl, n-pentyl, isopentyl, neopentyl, n-hexyl, 3-methylhexyl, 2,2-dimethylpentyl, 2,3-dimethylpentyl, 4,4-dimethylpentan-2-yl, n-heptyl, n-octyl, n-nonyl, n-decyl, undecyl, dodecyl, tridecyl, tetradecyl, pentadecyl, hexadecyl, heptadecyl, octadecyl, nonadecyl and icosyl.

[0053] The term “alkenyl,” as used herein, means a straight or branched hydrocarbon chain containing at least one carbon-carbon double bond. The double bond(s) may be located at any positions with the hydrocarbon chain. Representative examples of alkenyl include, but are not limited to, ethenyl, 2-propenyl, 2-methyl-2-propenyl, 3-butenyl, 4-pentenyl, 5-hexenyl, 2-heptenyl, 2-methyl-1-heptenyl, and 3-decenyl.

[0054] The term “alkynyl,” as used herein, means a straight or branched hydrocarbon chain

containing at least one carbon-carbon triple bond. The triple bond(s) may be located at any positions with the hydrocarbon chain. Representative examples of alkynyl include, but are not limited to, ethynyl, propynyl, and butynyl.

[0055] The term “amino,” as used herein, refers to an —NH₂ group. The term “alkylamino,” as used herein, refers to a group —NHR, wherein R is an alkyl group as defined herein. The term “dialkylamino,” as used herein, refers to a group —NR₂, wherein each R is independently an alkyl group as defined herein.

[0056] The term “cycloalkyl,” as used herein, refers to a saturated carbocyclic ring system containing three to ten carbon atoms and zero heteroatoms. The cycloalkyl may be monocyclic, bicyclic, bridged, fused, or spirocyclic. Representative examples of cycloalkyl include, but are not limited to, cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, cyclooctyl, cyclononyl, cyclodecyl, adamantyl, bicyclo[2.2.1]heptanyl, bicyclo[3.2.1]octanyl, and bicyclo[5.2.0]nonanyl.

[0057] The term “halogen” or “halo,” as used herein, means F, Cl, Br, or I.

[0058] The term “haloalkyl,” as used herein, means an alkyl group, as defined herein, in which at least one hydrogen atom (e.g., one, two, three, four, five, six, seven or eight hydrogen atoms) is replaced with a halogen. In some embodiments, each hydrogen atom of the alkyl group is replaced with a halogen. Representative examples of haloalkyl include, but are not limited to, fluoromethyl, difluoromethyl, trifluoromethyl, 2-fluoroethyl, 2,2,2-trifluoroethyl, and 3,3,3-trifluoropropyl.

[0059] The term “heteroalkyl,” as used herein, means an alkyl group, as defined herein, in which one or more of the carbon atoms (and any associated hydrogen atoms) are each independently replaced with a heteroatom group such as —NH—, —O—, —S—, —S(O)—, —S(O)₂—, —O—P(O)(O^{sup.}—)O—, or the like. By way of example, 1, 2, 3, 4, 5, 6, or more carbon atoms may be independently replaced with the same or different heteroatom group. A heteroalkyl group can also include one or more carbonyl moieties (i.e., wherein a carbon atom of the alkyl group is oxidized to a —C(O)— group).

[0060] The term “heteroalkenyl,” as used herein, refers to an alkenyl group, as defined herein, in which one or more of the carbon atoms (and any associated hydrogen atoms) are each independently replaced with a heteroatom group such as —NH—, —O—, —S—, —S(O)—, —S(O)₂—, or the like. By way of example, 1, 2, 3, 4, 5, 6, or more carbon atoms may be independently replaced with the same or different heteroatom group. A heteroalkenyl group can also include one or more carbonyl moieties (i.e., wherein a carbon atom of the alkyl group is oxidized to a —C(O)— group).

[0061] The term “heteroalkynyl,” as used herein, refers to an alkynyl group, as defined herein, in which one or more of the carbon atoms (and any associated hydrogen atoms) are each independently replaced with a heteroatom group such as —NH—, —O—, —S—, —S(O)—, —S(O)₂—, or the like. By way of example, 1, 2, 3, 4, 5, 6, or more carbon atoms may be independently replaced with the same or different heteroatom group. A heteroalkynyl group can also include one or more carbonyl moieties (i.e., wherein a carbon atom of the alkyl group is oxidized to a —C(O)— group).

[0062] The term “hydroxy,” as used herein, refers to an —OH group.

[0063] As used herein, the term “substituent” refers to a group substituted on an atom of the indicated group.

[0064] When a group or moiety can be substituted, the term “substituted” indicates that one or more (e.g., 1, 2, 3, 4, 5, or 6; in some embodiments 1, 2, or 3; and in other embodiments 1 or 2) hydrogen atoms on the group indicated in the expression using “substituted” can be replaced with a selection of recited indicated groups or with a suitable substituent group known to those of skill in the art (e.g., one or more of the groups recited below), provided that the designated atom's normal valence is not exceeded. Substituent groups include, but are not limited to, alkyl, alkenyl, alkynyl, alkoxy, acyl, amino, amido, amidino, aryl, azido, carbamoyl, carboxyl, carboxyl ester, cyano, cycloalkyl, cycloalkenyl, guanidino, halo, haloalkyl, haloalkoxy, heteroalkyl, heteroaryl,

heterocyclyl, hydroxy, hydrazino, imino, oxo, nitro, phosphate, phosphonate, sulfonic acid, sulfonamido, thiol, thione, thioxo, or combinations thereof.

[0065] As used herein, in chemical structures the indication:

##STR00008##

represents a point of attachment of one moiety to another moiety.

[0066] In some instances, the number of carbon atoms in a hydrocarbyl substituent (e.g., alkyl alkenyl) is indicated by the prefix “C.sub.x-C.sub.y”, wherein x is the minimum and y is the maximum number of carbon atoms in the substituent. Thus, for example, “C.sub.1-C.sub.3 alkyl” refers to an alkyl substituent containing from 1 to 3 carbon atoms.

[0067] For compounds described herein, groups and substituents thereof may be selected in accordance with permitted valence of the atoms and the substituents, such that the selections and substitutions result in a stable compound, e.g., which does not spontaneously undergo transformation such as by rearrangement, cyclization, elimination, etc.

[0068] Where substituent groups are specified by their conventional chemical formulae, written from left to right, they optionally encompass substituents resulting from writing the structure from right to left, e.g., —CH.sub.2O— is intended to encompass —OCH.sub.2—, and —C(O)NH— is intended to encompass —NHC(O)—.

[0069] Preferred methods and materials are described below, although methods and materials similar or equivalent to those described herein can be used in practice or testing of the present disclosure. All publications, patent applications, patents and other references mentioned herein are incorporated by reference in their entirety. The materials, methods, and examples disclosed herein are illustrative only and not intended to be limiting.

2. Compounds

[0070] In one aspect, disclosed is a compound of formula (I):

##STR00009## [0071] or a pharmaceutically acceptable salt thereof, wherein: [0072] R.sup.1 is hydrogen or C.sub.1-C.sub.6 alkyl; [0073] X is selected from O, NR.sup.2, S, and a bond; [0074] R.sup.2 is hydrogen, C.sub.1-C.sub.6 alkyl, C.sub.2-C.sub.6 alkenyl, C.sub.2-C.sub.6 alkynyl, C.sub.3-C.sub.6 cycloalkyl, or halo-C.sub.1-C.sub.6-alkyl; and [0075] L is selected from C.sub.8-C.sub.80 alkyl, C.sub.8-C.sub.80 alkenyl, C.sub.8-C.sub.80 alkynyl, C.sub.8-C.sub.80 heteroalkyl, C.sub.8-C.sub.80 heteroalkenyl, and C.sub.8-C.sub.80 heteroalkynyl, each of which is optionally substituted with one or more substituents selected from hydroxy and amino.

[0076] In some embodiments, R.sup.1 is hydrogen. In some embodiments, R.sup.1 is C.sub.1-C.sub.6 alkyl; In some embodiments, R' is methyl.

[0077] In some embodiments, X is O or a bond. In some embodiments, X is O. In some embodiments, X is a bond. In some embodiments, X is NR.sup.2, wherein R.sup.2 is selected from hydrogen and C.sub.1-C.sub.6 alkyl. In some embodiments, X is NH. In some embodiments, X is NR.sup.2, wherein R.sup.2 is methyl. In some embodiments, X is S.

[0078] In some embodiments, L is a lipid moiety having at least 8 carbon atoms. The lipid moiety can be derived from any suitable lipid, such as a fatty alcohol, a fatty acid, a phospholipid, a steroid, or a synthetic lipid. In some embodiments, the lipid moiety is derived from a lipid having a functional group, such as a hydroxy group, a carboxylic acid group, or an amino group, and the lipid moiety is attached to the compound of formula (I) via that functional group. For example, in some embodiments, L is a lipid moiety having at least 8 carbon atoms, and in such embodiments, the group X in formula (I) is derived from the functional group; for example, if the lipid moiety is derived from a fatty alcohol, X is O.

[0079] In some embodiments, L is selected from C.sub.8-C.sub.80 alkyl, C.sub.8-C.sub.80 alkenyl, C.sub.8-C.sub.80 alkynyl, C.sub.8-C.sub.80 heteroalkyl, C.sub.8-C.sub.80 heteroalkenyl, and C.sub.8-C.sub.80 heteroalkynyl, each of which is optionally substituted with one or more substituents selected from hydroxy and amino.

[0080] For example, in some embodiments, L is selected from C.sub.8-C.sub.80 alkyl and C.sub.8-

C.sub.80 alkenyl. In some embodiments, L is selected from C.sub.8-C.sub.40 alkyl and C.sub.8-C.sub.40 alkenyl. In some embodiments, the L is selected from C.sub.12-C.sub.40 alkyl and C.sub.12-C.sub.40 alkenyl. In some such embodiments, L is derived from a saturated or unsaturated fatty alcohol. In some embodiments, L is derived from linoleyl alcohol ((9Z,12Z)-octadeca-9,12-dien-1-ol), myristyl alcohol (1-tetradecanol), palmitoleyl alcohol ((Z)-hexadec-9-en-1-ol), oleyl alcohol ((Z)-octadec-9-en-1-ol), elaidyl alcohol (trans-9-octadecenol), cis-vaccenyl alcohol (cis-11-octadecenol), gadoleyl alcohol ((Z)-icos-9-en-1-ol), 11-eicosenol, erucyl alcohol (cis-13-docosenol), 15-tetracosen-1-ol, eicosadienyl alcohol (icosa-11,14-dien-1-ol), linolenyl alcohol ((9Z,12Z,15Z)-9,12,15-octadecatrien-1-ol), γ -linolenyl alcohol ((6E,9E,12E)-octadeca-6,9,12-trien-1-ol), eleostearyl alcohol (octadeca-9,11,13-trien-1-ol), icosa-5,8,11-trien-1-ol, eicos-13-en-1-ol, icosa-11,14-17-trien-1-ol, octadeca-6,9,12,15-tetraen-1-ol, arachidonyl alcohol ((5Z,8Z,11Z,14Z)-icosatetraen-1-ol), 4E,6Z-hexadecadien-1-ol, icosa-5,8,11,14,17-pentaen-1-ol, docosahexaenoyl alcohol (docosa-4,7,10,13,16,19-hexaen-1-ol), docosa-7,10,13,16,19-pentaen-1-ol, tetracos-6,9,12,15,18,21-hexaen-1-ol, capryl alcohol (1-octanol), pelargonic alcohol (1-nonanol), decyl alcohol (1-decanol), undecyl alcohol (1-undecanol), lauryl alcohol (1-dodecanol), tridecyl alcohol (1-tridecanol), myristyl alcohol (1-tetradecanol), pentadecyl alcohol (1-pentadecanol), cetyl alcohol (1-hexadecanol), palmitoleyl alcohol (cis-9-hexadecen-1-ol), heptadecylalcohol (1-n-heptadecanol), stearyl alcohol (1-octadecanol), oleyl alcohol (1-octadecenol), nonadecyl alcohol (1-nonadecanol), arachidyl alcohol (1-eicosanol), heneicosyl alcohol (1-heneicosanol), behenyl alcohol (1-docosanol), erucyl alcohol (cis-13-docosen-1-ol), 1-tricosanol, lignoceryl alcohol (1-tetracosanol), pentacosylic alcohol (1-pentacosanol), ceryl alcohol (1-hexacosanol), 1-heptacosanol, montanyl alcohol (1-octacosanol), 1-nonacosanol, myricyl alcohol (1-triacontanol), 1-hentriacontanol, 1-dotriacontanol (lacceryl alcohol), 1-tritriacontanol, geddyl alcohol (1-tetratriacontanol), 1-hexatriacontanol, 1-heptatriacontanol, 1-octatriacontanol, nonatriacontan-1-ol, or 1-tetracontanol.

[0081] In some embodiments, L is selected from C.sub.8-C.sub.80 heteroalkyl, C.sub.8-C.sub.80 heteroalkenyl, and C.sub.8-C.sub.80 heteroalkynyl. In such embodiments, L can be derived from a lipid including one or more heteroatom groups, such as —O—, —NH—, —C(O)—, or the like, or combinations thereof (e.g., —C(O)O— groups).

[0082] For example, in some embodiments, the L has a formula (A):

##STR00010## [0083] wherein: [0084] n is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, or 40; and

[0085] R.sup.a and R.sup.b are each independently selected from C.sub.6-C.sub.40 alkyl, C.sub.6-C.sub.40 alkenyl, C.sub.6-C.sub.40 heteroalkyl, and C.sub.6-C.sub.40 heteroalkenyl.

[0086] For example, in some embodiments, n is 1, 2, 3, 4, 5, 6, 7, or 8. In some embodiments, n is 2, 3, or 4. In some embodiments, n is 2. In some embodiments, n is 3. In some embodiments, n is 4.

[0087] In some embodiments, R.sup.a and R.sup.b are each independently selected from C.sub.6-C.sub.40 alkyl and C.sub.6-C.sub.40 alkenyl. For example, in some embodiments, R.sup.a and R.sup.b are each independently selected from n-hexyl, n-heptyl, n-octyl, n-nonyl, n-decyl, undecyl, dodecyl, tridecyl, tetradecyl, pentadecyl, hexadecyl, heptadecyl, octadecyl, nonadecyl, icosyl, heneicosyl, docosyl, tricosyl, tetracosyl, pentacosyl, hexacosyl, heptacosyl, octacosyl, nonacosyl, triacontyl, hentriacontyl, dotriacontyl, tritriacontyl, tetratriacontyl, pentatriacontyl, hexatriacontyl, heptatriacontyl, octatriacontyl, nonatriacontyl, tetracontyl, linoleyl((9Z,12Z)-octadeca-9,12-dien-1-yl), palmitoleyl((Z)-hexadec-9-en-1-yl), oleyl((Z)-octadec-9-en-1-yl), elaidyl(trans-9-octadecenyl), cis-vaccenyl(cis-11-octadecenyl), gadoleyl((Z)-icos-9-en-1-yl), 11-eicosenyl, erucyl(cis-13-docosenyl), 15-tetracosen-1-yl, eicosadienyl(icosa-11,14-dien-1-yl), linolenyl((9Z,12Z,15Z)-9,12,15-octadecatrien-1-yl), γ -linolenyl((6E,9E,12E)-octadeca-6,9,12-trien-1-yl), eleostearyl(octadeca-9,11,13-trien-1-yl), icosa-5,8,11-trien-1-yl, eicos-13-en-1-yl, icosa-11,14-17-trien-1-yl, octadeca-6,9,12,15-tetraen-1-yl, arachidonyl((5Z,8Z,11Z,14Z)-icosatetraen-1-yl), 4E,6Z-hexadecadien-1-yl, icosa-5,8,11,14,17-pentaen-1-yl,

docosaheptaenyl(docosa-4,7,10,13,16,19-hexaen-1-yl), docosa-7,10,13,16,19-pentaen-1-yl, and tetracosahexaenyl(tetracos-6,9,12,15,18,21-hexaen-1-yl). In some embodiments, R.sup.a and R.sup.b are each linoleyl.

[0088] In some embodiments, L has a formula (B) or (C):

##STR00011## [0089] wherein: [0090] n and p are each independently 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, or 40; [0091] R.sup.a1 is selected from C.sub.6-C.sub.40 alkyl and C.sub.6-C.sub.40 alkenyl; and [0092] R.sup.b is selected from C.sub.6-C.sub.40 alkyl, C.sub.6-C.sub.40 alkenyl, C.sub.6-C.sub.40 heteroalkyl, and C.sub.6-C.sub.40 heteroalkenyl.

[0093] In some embodiments, n is 1, 2, 3, 4, 5, 6, 7, or 8. In some embodiments, n is 2, 3, or 4. In some embodiments, n is 2. In some embodiments, n is 3. In some embodiments, n is 4.

[0094] In some embodiments, p is 1, 2, 3, 4, 5, 6, 7, or 8. In some embodiments, p is 3, 4, 5, 6, or 7. In some embodiments, p is 3. In some embodiments, p is 4. In some embodiments, p is 5. In some embodiments, p is 6. In some embodiments, p is 7.

[0095] In some embodiments, R.sup.a1 is C.sub.6-C.sub.24 alkyl or C.sub.6-C.sub.24 alkenyl. In some embodiments, R.sup.a1 is selected from C.sub.9-C.sub.22 alkyl and C.sub.9-C.sub.22 alkenyl. In some embodiments, R.sup.a1 is selected from straight or branched C.sub.6, C.sub.7, C.sub.8, C.sub.9, C.sub.10, C.sub.11, C.sub.12, C.sub.13, C.sub.14, C.sub.15, C.sub.16, C.sub.17, C.sub.18, C.sub.19, C.sub.20, C.sub.21, C.sub.22, C.sub.23, or C.sub.24 alkyl. In some embodiments, R.sup.a1 is selected from straight or branched C.sub.6, C.sub.7, C.sub.8, C.sub.9, C.sub.10, C.sub.11, C.sub.12, C.sub.13, C.sub.14, C.sub.15, C.sub.16, C.sub.17, C.sub.18, C.sub.19, C.sub.20, C.sub.21, C.sub.22, C.sub.23, or C.sub.24 alkenyl. In some embodiments, R.sup.a1 is selected from linoleyl, n-nonyl, n-undecyl, heneicosan-11-yl, pentadecane-7-yl, and heptadecan-9-yl.

[0096] In some embodiments, R.sup.b is selected from C.sub.6-C.sub.40 alkyl and C.sub.6-C.sub.40 alkenyl. For example, in some embodiments, R.sup.b is selected from n-hexyl, n-heptyl, n-octyl, n-nonyl, n-decyl, undecyl, dodecyl, tridecyl, tetradecyl, pentadecyl, hexadecyl, heptadecyl, octadecyl, nonadecyl, icosyl, heneicosyl, docosyl, tricosyl, tetracosyl, pentacosyl, hexacosyl, heptacosyl, octacosyl, nonacosyl, triacontyl, hentriacontyl, dotriacontyl, tritriacontyl, tetratriacontyl, pentatriacontyl, hexatriacontyl, heptatriacontyl, octatriacontyl, nonatriacontyl, tetracontyl, linoleyl((9Z,12Z)-octadeca-9,12-dien-1-yl), palmitoleyl((Z)-hexadec-9-en-1-yl), oleyl((Z)-octadec-9-en-1-yl), elaidyl(trans-9-octadecenyl), cis-vaccenyl(cis-11-octadecenyl), gadoleyl((Z)-icos-9-en-1-yl), 11-eicosenyl, erucyl(cis-13-docosenyl), 15-tetracosen-1-yl, eicosadienyl(icos-11,14-dien-1-yl), linolenyl((9Z,12Z,15Z)-9,12,15-octadecatrien-1-yl), γ -linolenyl((6E,9E,12E)-octadeca-6,9,12-trien-1-yl), eleostearyl (octadeca-9,11,13-trien-1-yl), icos-5,8,11-trien-1-yl, eicos-13-en-1-yl, icos-11,14-17-trien-1-yl, octadeca-6,9,12,15-tetraen-1-yl, arachidonyl((5Z,8Z,11Z,14Z)-icosatetraen-1-yl), 4E,6Z-hexadecadien-1-yl, icos-5,8,11,14,17-pentaen-1-yl, docosaheptaenyl(docosa-4,7,10,13,16,19-hexaen-1-yl), docosa-7,10,13,16,19-pentaen-1-yl, and tetracosahexaenyl(tetracos-6,9,12,15,18,21-hexaen-1-yl). In some embodiments, R.sup.b is linoleyl.

[0097] In some embodiments, the L has a formula (D), (E), or (F):

##STR00012## [0098] wherein: [0099] n, p, and q are each independently 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, or 40; and [0100] R.sup.a1 and R.sup.a2 are each independently selected from C.sub.6-C.sub.40 alkyl and C.sub.6-C.sub.40 alkenyl.

[0101] In some embodiments, L has formula (D). In some embodiments, L has formula (E). In some embodiments, L has formula (F). In some embodiments, n is 1, 2, 3, 4, 5, 6, 7, or 8. In some embodiments, n is 2, 3, or 4. In some embodiments, n is 2. In some embodiments, n is 3. In some embodiments, n is 4.

[0102] In some embodiments, p and q are each independently selected from 1, 2, 3, 4, 5, 6, 7, or 8. In some embodiments, p and q are each independently selected from 3, 4, 5, 6, and 7. In some

embodiments, p and q are each independently selected from 3, 5, and 7. In some embodiments, p and q are each 5. In some embodiments, p and q are each 6. In some embodiments, p and q are each 7.

[0103] In some embodiments, R^{sup.a1} and R^{sup.a2} are each independently selected from C_{sub.6}-C_{sub.24} alkyl and C_{sub.6}-C_{sub.24} alkenyl. In some embodiments, R^{sup.a1} and R^{sup.a2} are each independently selected from C_{sub.9}-C_{sub.22} alkyl and C_{sub.9}-C_{sub.22} alkenyl. In some embodiments, R^{sup.a1} and R^{sup.a2} are each independently selected from straight or branched C_{sub.6}, C_{sub.7}, C_{sub.8}, C_{sub.9}, C_{sub.10}, C_{sub.11}, C_{sub.12}, C_{sub.13}, C_{sub.14}, C_{sub.15}, C_{sub.16}, C_{sub.17}, C_{sub.18}, C_{sub.19}, C_{sub.20}, C_{sub.21}, C_{sub.22}, C_{sub.23}, or C_{sub.24} alkyl. In some embodiments, R^{sup.a1} and R^{sup.a2} are each independently selected from straight or branched C_{sub.6}, C_{sub.7}, C_{sub.8}, C_{sub.9}, C_{sub.10}, C_{sub.11}, C_{sub.12}, C_{sub.13}, C_{sub.14}, C_{sub.15}, C_{sub.16}, C_{sub.17}, C_{sub.18}, C_{sub.19}, C_{sub.20}, C_{sub.21}, C_{sub.22}, C_{sub.23}, or C_{sub.24} alkenyl. In some embodiments, R^{sup.a1} and R^{sup.a2} are each independently selected from linoleyl, n-nonyl, n-undecyl, heneicosan-11-yl, pentadecane-7-yl, and heptadecan-9-yl.

[0104] In some embodiments, L is derived from a steroid. For example, in some embodiments, L is derived from cholesterol, beta-sisterol, or BHEM-cholesterol.

[0105] In some embodiments, L is selected from:

##STR00013## ##STR00014## ##STR00015##

[0106] The compounds may exist as a stereoisomer wherein asymmetric or chiral centers are present. The stereoisomer is "R" or "S" depending on the configuration of substituents around the chiral carbon atom. The terms "R" and "S" used herein are configurations as defined in IUPAC 1974 Recommendations for Section E, Fundamental Stereochemistry, in Pure Appl. Chem., 1976, 45:13-30. The disclosure contemplates various stereoisomers and mixtures thereof and these are specifically included within the scope of this disclosure. Stereoisomers include enantiomers and diastereomers, and mixtures of enantiomers or diastereomers. Individual stereoisomers of the compounds may be prepared synthetically from commercially available starting materials, which contain asymmetric or chiral centers or by preparation of racemic mixtures followed by methods of resolution well-known to those of ordinary skill in the art. These methods of resolution are exemplified by (1) attachment of a mixture of enantiomers to a chiral auxiliary, separation of the resulting mixture of diastereomers by recrystallization or chromatography and optional liberation of the optically pure product from the auxiliary as described in Furniss, Hannaford, Smith, and Tatchell, "Vogel's Textbook of Practical Organic Chemistry," 5th edition (1989), Longman Scientific & Technical, Essex CM20 2JE, England (or more recent versions thereof), or (2) direct separation of the mixture of optical enantiomers on chiral chromatographic columns, or (3) fractional recrystallization methods.

[0107] It should be understood that the compounds may possess tautomeric forms, as well as geometric isomers, and that these also constitute embodiments of the disclosure.

[0108] The present disclosure also includes isotopically-labeled compounds, which is identical to those recited in formula (I), but for the fact that one or more atoms are replaced by an atom having an atomic mass or mass number different from the atomic mass or mass number usually found in nature. Examples of isotopes include those for hydrogen, carbon, nitrogen, oxygen, phosphorus, sulfur, fluorine, and chlorine, such as, but not limited to ²H, ³H, ¹³C, ¹⁴C, ¹⁵N, ¹⁸O, ¹⁷O, ³¹P, ³²P, ³⁵S, ¹⁸F, and ³⁶Cl, respectively. Substitution with heavier isotopes such as deuterium, for example, ²H, can afford certain therapeutic advantages resulting from greater metabolic stability, for example increased in vivo half-life or reduced dosage requirements and, hence, may be preferred in some circumstances. The compound may incorporate positron-emitting isotopes for medical imaging and positron-emitting tomography (PET) studies for determining the distribution of receptors. Suitable positron-emitting isotopes that can be incorporated in compounds of formula (I) are ¹C, ¹³N, ¹⁵O, and

.sup.18F. Isotopically-labeled compounds of formula (I) can generally be prepared by conventional techniques known to those skilled in the art or by processes analogous to those described in the accompanying Examples using appropriate isotopically-labeled reagent in place of non-isotopically-labeled reagent.

[0109] The disclosed compounds may exist as pharmaceutically acceptable salts. The term “pharmaceutically acceptable salt” refers to salts or zwitterions of the compounds which are water or oil-soluble or dispersible, suitable for treatment of disorders without undue toxicity, irritation, and allergic response, commensurate with a reasonable benefit/risk ratio and effective for their intended use. The salts may be prepared during the final isolation and purification of the compounds or separately by reacting an amino group of the compounds with a suitable acid. For example, a compound may be dissolved in a suitable solvent, such as but not limited to methanol and water and treated with at least one equivalent of an acid, like hydrochloric acid. The resulting salt may precipitate out and be isolated by filtration and dried under reduced pressure. Alternatively, the solvent and excess acid may be removed under reduced pressure to provide a salt. Representative salts include acetate, adipate, alginate, citrate, aspartate, benzoate, benzenesulfonate, bisulfate, butyrate, camphorate, camphorsulfonate, digluconate, glycerophosphate, hemisulfate, heptanoate, hexanoate, formate, isethionate, fumarate, lactate, maleate, methanesulfonate, naphthylenesulfonate, nicotinate, oxalate, pamoate, pectinate, persulfate, 3-phenylpropionate, picrate, oxalate, maleate, pivalate, propionate, succinate, tartrate, trichloroacetate, trifluoroacetate, glutamate, para-toluenesulfonate, undecanoate, hydrochloric, hydrobromic, sulfuric, phosphoric and the like. The amino groups of the compounds may also be quaternized with alkyl chlorides, bromides, and iodides such as methyl, ethyl, propyl, isopropyl, butyl, lauryl, myristyl, stearyl and the like.

[0110] Basic addition salts may be prepared during the final isolation and purification of the disclosed compounds by reaction of a carboxyl group with a suitable base such as the hydroxide, carbonate, or bicarbonate of a metal cation such as lithium, sodium, potassium, calcium, magnesium, or aluminum, or an organic primary, secondary, or tertiary amine. Quaternary amine salts can be prepared, such as those derived from methylamine, dimethylamine, trimethylamine, triethylamine, diethylamine, ethylamine, tributylamine, pyridine, N,N-dimethylaniline, N-methylpiperidine, N-methylmorpholine, dicyclohexylamine, procaine, dibenzylamine, N,N-dibenzylpbenethylamine, 1-ephedrine and N,N'-dibenzylethylenediamine, ethylenediamine, ethanolamine, diethanolamine, piperidine, piperazine, and the like.

[0111] Compounds may be synthesized according to a variety of methods, including those illustrated in the Examples. Reaction conditions and reaction times for each individual step can vary depending on the particular reactants employed and substituents present in the reactants used. Specific procedures are provided in the Examples section. Reactions can be worked up in the conventional manner, e.g., by eliminating the solvent from the residue and further purified according to methodologies generally known in the art such as, but not limited to, crystallization, distillation, extraction, trituration, and chromatography. Unless otherwise described, the starting materials and reagents are either commercially available or can be prepared by one skilled in the art from commercially available materials using methods described in the chemical literature. Starting materials, if not commercially available, can be prepared by procedures selected from standard organic chemical techniques, techniques that are analogous to the synthesis of known, structurally similar compounds, or techniques that are analogous to the above described schemes or the procedures described in the synthetic examples section.

[0112] Routine experimentations, including appropriate manipulation of the reaction conditions, reagents and sequence of the synthetic route, protection of any chemical functionality that cannot be compatible with the reaction conditions, and deprotection at a suitable point in the reaction sequence of the method are included in the scope of the disclosure. Suitable protecting groups and the methods for protecting and deprotecting different substituents using such suitable protecting

groups are well known to those skilled in the art; examples of which can be found in PGM Wuts and TW Greene, in Greene's book titled Protective Groups in Organic Synthesis (4th ed.), John Wiley & Sons, NY (2006), which is incorporated herein by reference in its entirety. Synthesis of the compounds of the disclosure can be accomplished by methods analogous to those described in the synthetic schemes described hereinabove and in specific examples.

[0113] When an optically active form of a disclosed compound is required, it can be obtained by carrying out one of the procedures described herein using an optically active starting material (prepared, for example, by asymmetric induction of a suitable reaction step), or by resolution of a mixture of the stereoisomers of the compound or intermediates using a standard procedure (such as chromatographic separation, recrystallization, or enzymatic resolution).

[0114] Similarly, when a pure geometric isomer of a compound is required, it can be obtained by carrying out one of the above procedures using a pure geometric isomer as a starting material, or by resolution of a mixture of the geometric isomers of the compound or intermediates using a standard procedure such as chromatographic separation.

[0115] It can be appreciated that the synthetic schemes and specific examples as described are illustrative and are not to be read as limiting the scope of the disclosure as it is defined in the appended claims. All alternatives, modifications, and equivalents of the synthetic methods and specific examples are included within the scope of the claims.

3. Compositions

[0116] The disclosure also provides lipid compositions that may be suitable for administration to a subject (such as a patient, which may be a human or non-human) comprising a compound of formula (II):

W—X-L (II) [0117] or a pharmaceutically acceptable salt thereof, wherein: [0118] W is an amino acid; [0119] X is selected from O, NR^{sup.2}, S, and a bond; [0120] R^{sup.2} is hydrogen C_{sub.1}-C_{sub.6} alkyl, C_{sub.2}-C_{sub.6} alkenyl, C_{sub.2}-C_{sub.6} alkynyl, C_{sub.3}-C_{sub.6} cycloalkyl, or halo-C_{sub.1}-C_{sub.6}-alkyl; and [0121] L is selected from C_{sub.8}-C_{sub.80} alkyl, C_{sub.8}-C_{sub.80} alkenyl, C_{sub.8}-C_{sub.80} alkynyl, C_{sub.8}-C_{sub.80} heteroalkyl, C_{sub.8}-C_{sub.80} heteroalkenyl, and C_{sub.8}-C_{sub.80} heteroalkynyl, each of which is optionally substituted with one or more substituents selected from hydroxy and amino.

[0122] The term “amino acid” as used here refers to any and all amino acids, including naturally occurring amino acids (e.g., α-amino acids), unnatural amino acids, modified amino acids, and non-natural amino acids. In some embodiments, the term “amino acid” refers to any molecule that contains both amine and carboxyl terminal functional groups. In some embodiments, the term “amino acid” refers to all amino acid-like compounds that are similar in structure and/or overall shape to naturally occurring amino acids. Amino acids can be D- or L-amino acids. Natural amino acids include those found in nature, such as, e.g., the amino acids that combine into peptide chains to form the building-blocks of a vast array of proteins. These are primarily L stereoisomers, although a few D-amino acids occur in bacterial envelopes and some antibiotics. The “non-standard,” natural amino acids include, for example, pyrrolysine (found in methanogenic organisms and other eukaryotes), selenocysteine (present in many non-eukaryotes as well as most eukaryotes), and N-formylmethionine (encoded by the start codon AUG in bacteria, mitochondria, and chloroplasts). “Unnatural” or “non-natural” amino acids are non-proteinogenic amino acids (e.g., those not naturally encoded or found in the genetic code) that either occur naturally or are chemically synthesized. Over 140 unnatural amino acids are known and thousands of more combinations are possible. Examples of “unnatural” amino acids include β-amino acids (β^{sup.3} and β^{sup.2}), homo-amino acids, proline and pyruvic acid derivatives, 3-substituted alanine derivatives, glycine derivatives, ring-substituted phenylalanine and tyrosine derivatives, linear core amino acids, diamino acids, D-amino acids, α-methyl amino acids and N-methyl amino acids. Unnatural or non-natural amino acids also include modified amino acids. “Modified” amino acids

include amino acids (e.g., natural amino acids) that have been chemically modified to include a group, groups, or chemical moiety not naturally present on the amino acid.

[0123] Reference to a particular amino acid is meant to encompass that amino acid in both its conjugated and non-conjugated state. Thus, the term “amino acid” refers to a residue of an amino acid in which a portion of its structure participates in a bond, often resulting in loss of atoms from the amino acid itself, for example, a loss of the equivalent of a water molecule when forming a peptide bond in a protein. The amino acid may be conjugated to the lipid moiety by any suitable functional group within the amino acid. In some embodiments, the amino acid is conjugated at its carboxy terminus. In some embodiments, the amino acid is conjugated at the carbonyl carbon atom of the carboxylic acid residue.

[0124] For the most part, the names of naturally occurring and non-naturally occurring aminoacyl residues used herein follow the naming conventions suggested by the IUPAC Commission on the Nomenclature of Organic Chemistry and the IUPAC-IUB Commission on Biochemical Nomenclature as set out in “Nomenclature of α -Amino Acids (Recommendations, 1974)” Biochemistry, 14 (2), (1975). To the extent that the names and abbreviations of amino acids and aminoacyl residues employed in this specification and appended claims differ from those suggestions, they will be made clear to the reader.

[0125] Throughout the present specification, unless naturally occurring amino acids are referred to by their full name (e.g., alanine, arginine, etc.), they are designated by their conventional three-letter or single-letter abbreviations (e.g., Ala or A for alanine, Arg or R for arginine, etc.). The term “L-amino acid,” as used herein, refers to the “L” isomeric form of a peptide, and conversely the term “D-amino acid” refers to the “D” isomeric form of a peptide (e.g., Dphe, (D) Phe, D-Phe, or .sup.DF for the D isomeric form of Phenylalanine). Amino acid residues in the D isomeric form can be substituted for any L-amino acid residue, as long as the desired function is retained.

[0126] In the case of less common or non-naturally occurring amino acids, unless they are referred to by their full name (e.g., sarcosine, ornithine, etc.), frequently employed three- or four-character codes are employed for residues thereof, including, Sar or Sarc (sarcosine, e.g., N-methylglycine), Aib (α -aminoisobutyric acid), Dab (2,4-diaminobutanoic acid), Dapa (2,3-diaminopropanoic acid), γ -Glu (γ -glutamic acid), Gaba (γ -aminobutanoic acid), β -Pro (pyrrolidine-3-carboxylic acid), and 8Ado (8-amino-3,6-dioxaoctanoic acid), Abu (2-amino butyric acid), β hPro (β -homoproline), β hPhe (β -homophenylalanine) and Bip (β,β diphenylalanine), and Ida (Iminodiacetic acid).

[0127] In some embodiments, W is tryptophan, proline, lysine, histidine, arginine, or a variant thereof. In some embodiments, W is tryptophan or a variant thereof.

[0128] In some embodiments, the compound formula (II) is a compound of formula (IIa), wherein W is 1-methyl-D-tryptophan. In some embodiments, the compound of formula (IIa) is:

##STR00016## [0129] or a pharmaceutically acceptable salt thereof.

[0130] In some embodiments, X is O or a bond. In some embodiments, X is O. In some embodiments, X is a bond. In some embodiments, X is NR.sup.2, wherein R.sup.2 is selected from hydrogen and C.sub.1-C.sub.6 alkyl. In some embodiments, X is NH. In some embodiments, X is NR.sup.2, wherein R.sup.2 is methyl. In some embodiments, X is S.

[0131] In some embodiments, L is a lipid moiety having at least 8 carbon atoms. The lipid moiety can be derived from any suitable lipid, such as a fatty alcohol, a fatty acid, a phospholipid, a steroid, or a synthetic lipid. In some embodiments, the lipid moiety is derived from a lipid having a functional group, such as a hydroxy group, a carboxylic acid group, or an amino group, and the lipid moiety is attached to the compound of formula (II) via that functional group. For example, in some embodiments, L is a lipid moiety having at least 8 carbon atoms, and in such embodiments, the group X in formula (II) is derived from the functional group; for example, if the lipid moiety is derived from a fatty alcohol, X is O.

[0132] In some embodiments, L is selected from C.sub.8-C.sub.80 alkyl, C.sub.8-C.sub.80 alkenyl, C.sub.8-C.sub.80 alkynyl, C.sub.8-C.sub.80 heteroalkyl, C.sub.8-C.sub.80 heteroalkenyl, and

C.sub.8-C.sub.80 heteroalkynyl, each of which is optionally substituted with one or more substituents selected from hydroxy and amino.

[0133] For example, in some embodiments, L is selected from C.sub.8-C.sub.80 alkyl and C.sub.8-C.sub.80 alkenyl. In some embodiments, L is selected from C.sub.8-C.sub.40 alkyl and C.sub.8-C.sub.40 alkenyl. In some embodiments, the L is selected from C.sub.12-C.sub.40 alkyl and C.sub.12-C.sub.40 alkenyl. In some such embodiments, L is derived from a saturated or unsaturated fatty alcohol. In some embodiments, L is derived from linoleyl alcohol ((9Z,12Z)-octadeca-9,12-dien-1-ol), myristyl alcohol (1-tetradecanol), palmitoleyl alcohol ((Z)-hexadec-9-en-1-ol), oleyl alcohol ((Z)-octadec-9-en-1-ol), elaidyl alcohol (trans-9-octadecenol), cis-vaccenyl alcohol (cis-11-octadecenol), gadoleyl alcohol ((Z)-icos-9-en-1-ol), 11-eicosenol, erucyl alcohol (cis-13-docosenol), 15-tetracosen-1-ol, eicosadienyl alcohol (icosa-11,14-dien-1-ol), linolenyl alcohol ((9Z,12Z,15Z)-9,12,15-octadecatrien-1-ol), γ -linolenyl alcohol ((6E,9E,12E)-octadeca-6,9,12-trien-1-ol), eleostearyl alcohol (octadeca-9,11,13-trien-1-ol), icosa-5,8,11-trien-1-ol, eicos-13-en-1-ol, icosa-11,14-17-trien-1-ol, octadeca-6,9,12,15-tetraen-1-ol, arachidonyl alcohol ((5Z,8Z,11Z,14Z)-icosatetraen-1-ol), 4E,6Z-hexadecadien-1-ol, icosa-5,8,11,14,17-pentaen-1-ol, docosahexaenoyl alcohol (docosa-4,7,10,13,16,19-hexaen-1-ol), docosa-7,10,13,16,19-pentaen-1-ol, tetracos-6,9,12,15,18,21-hexaen-1-ol, capryl alcohol (1-octanol), pelargonic alcohol (1-nonanol), decyl alcohol (1-decanol), undecyl alcohol (1-undecanol), lauryl alcohol (1-dodecanol), tridecyl alcohol (1-tridecanol), myristyl alcohol (1-tetradecanol), pentadecyl alcohol (1-pentadecanol), cetyl alcohol (1-hexadecanol), palmitoleyl alcohol (cis-9-hexadecen-1-ol), heptadecylalcohol (1-n-heptadecanol), stearyl alcohol (1-octadecanol), oleyl alcohol (1-octadecenol), nonadecyl alcohol (1-nonadecanol), arachidyl alcohol (1-eicosanol), heneicosyl alcohol (1-heneicosanol), behenyl alcohol (1-docosanol), erucyl alcohol (cis-13-docosen-1-ol), 1-tricosanol, lignoceryl alcohol (1-tetracosanol), pentacosylic alcohol (1-pentacosanol), ceryl alcohol (1-hexacosanol), 1-heptacosanol, montanyl alcohol (1-octacosanol), 1-nonacosanol, myricyl alcohol (1-triacontanol), 1-hentriacontanol, 1-dotriacontanol (lacceryl alcohol), 1-tritriacontanol, geddyl alcohol (1-tetratriacontanol), 1-hexatriacontanol, 1-heptatriacontanol, 1-octatriacontanol, nonatriacontan-1-ol, or 1-tetracontanol.

[0134] In some embodiments, L is selected from C.sub.8-C.sub.80 heteroalkyl, C.sub.8-C.sub.80 heteroalkenyl, and C.sub.8-C.sub.80 heteroalkynyl. In such embodiments, L can be derived from a lipid including one or more heteroatom groups, such as —O—, —NH—, —C(O)—, or the like, or combinations thereof (e.g., —C(O)O— groups).

[0135] For example, in some embodiments, the L has a formula (A):

##STR00017## [0136] wherein: [0137] n is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, or 40; and

[0138] R.sup.a and R.sup.b are each independently selected from C.sub.6-C.sub.40 alkyl, C.sub.6-C.sub.40 alkenyl, C.sub.6-C.sub.40 heteroalkyl, and C.sub.6-C.sub.40 heteroalkenyl.

[0139] For example, in some embodiments, n is 1, 2, 3, 4, 5, 6, 7, or 8. In some embodiments, n is 2, 3, or 4. In some embodiments, n is 2. In some embodiments, n is 3. In some embodiments, n is 4.

[0140] In some embodiments, R.sup.a and R.sup.b are each independently selected from C.sub.6-C.sub.40 alkyl and C.sub.6-C.sub.40 alkenyl. For example, in some embodiments, R.sup.a and R.sup.b are each independently selected from n-hexyl, n-heptyl, n-octyl, n-nonyl, n-decyl, undecyl, dodecyl, tridecyl, tetradecyl, pentadecyl, hexadecyl, heptadecyl, octadecyl, nonadecyl, icosyl, henicosyl, docosyl, tricosyl, tetracosyl, pentacosyl, hexacosyl, heptacosyl, octacosyl, nonacosyl, triacontyl, hentriacontyl, dotriacontyl, tritriacontyl, tetratriacontyl, pentatriacontyl, hexatriacontyl, heptatriacontyl, octatriacontyl, nonatriacontyl, tetracontyl, linoleyl((9Z,12Z)-octadeca-9,12-dien-1-yl), palmitoleyl((Z)-hexadec-9-en-1-yl), oleyl((Z)-octadec-9-en-1-yl), elaidyl(trans-9-octadecenyl), cis-vaccenyl(cis-11-octadecenyl), gadoleyl((Z)-icos-9-en-1-yl), 11-eicosenyl, erucyl(cis-13-docosenyl), 15-tetracosen-1-yl, eicosadienyl(icosa-11,14-dien-1-yl), linolenyl((9Z,12Z,15Z)-9,12,15-octadecatrien-1-yl), γ -linolenyl((6E,9E,12E)-octadeca-6,9,12-

trien-1-yl), eleostearyl(octadeca-9,11,13-trien-1-yl), icoso-5,8,11-trien-1-yl, eicos-13-en-1-yl, icoso-11,14-17-trien-1-yl, octadeca-6,9,12,15-tetraen-1-yl, arachidonyl((5Z,8Z,11Z,14Z)-icosatetraen-1-yl), 4E,6Z-hexadecadien-1-yl, icoso-5,8,11,14,17-pentaen-1-yl, docosahexaenoyl(docosa-4,7,10,13,16,19-hexaen-1-yl), docosa-7,10,13,16,19-pentaen-1-yl, and tetracosa-6,9,12,15,18,21-hexaen-1-yl. In some embodiments, R.sup.a and R.sup.b are each linoleyl.

[0141] In some embodiments, L has a formula (B) or (C):

##STR00018## [0142] wherein: [0143] n and p are each independently 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, or 40; [0144] R.sup.a1 is selected from C.sub.6-C.sub.40 alkyl and C.sub.6-C.sub.40 alkenyl; and [0145] R.sup.b is selected from C.sub.6-C.sub.40 alkyl, C.sub.6-C.sub.40 alkenyl, C.sub.6-C.sub.40 heteroalkyl, and C.sub.6-C.sub.40 heteroalkenyl.

[0146] In some embodiments, n is 1, 2, 3, 4, 5, 6, 7, or 8. In some embodiments, n is 2, 3, or 4. In some embodiments, n is 2. In some embodiments, n is 3. In some embodiments, n is 4.

[0147] In some embodiments, p is 1, 2, 3, 4, 5, 6, 7, or 8. In some embodiments, p is 3, 4, 5, 6, or 7. In some embodiments, p is 3. In some embodiments, p is 4. In some embodiments, p is 5. In some embodiments, p is 6. In some embodiments, p is 7.

[0148] In some embodiments, R.sup.a1 is C.sub.6-C.sub.24 alkyl or C.sub.6-C.sub.24 alkenyl. In some embodiments, R.sup.a1 is selected from C.sub.9-C.sub.22 alkyl and C.sub.9-C.sub.22 alkenyl. In some embodiments, R.sup.a1 is selected from straight or branched C.sub.6, C.sub.7, C.sub.8, C.sub.9, C.sub.10, C.sub.11, C.sub.12, C.sub.13, C.sub.14, C.sub.15, C.sub.16, C.sub.17, C.sub.18, C.sub.19, C.sub.20, C.sub.21, C.sub.22, C.sub.23, or C.sub.24 alkyl. In some embodiments, R.sup.a1 is selected from straight or branched C.sub.6, C.sub.7, C.sub.8, C.sub.9, C.sub.10, C.sub.11, C.sub.12, C.sub.13, C.sub.14, C.sub.15, C.sub.16, C.sub.17, C.sub.18, C.sub.19, C.sub.20, C.sub.21, C.sub.22, C.sub.23, or C.sub.24 alkenyl. In some embodiments, R.sup.a1 is selected from linoleyl, n-nonyl, n-undecyl, heneicosan-11-yl, pentadecane-7-yl, and heptadecan-9-yl.

[0149] In some embodiments, R' is selected from C.sub.6-C.sub.40 alkyl and C.sub.6-C.sub.40 alkenyl. For example, in some embodiments, RD is selected from n-hexyl, n-heptyl, n-octyl, n-nonyl, n-decyl, undecyl, dodecyl, tridecyl, tetradecyl, pentadecyl, hexadecyl, heptadecyl, octadecyl, nonadecyl, icosyl, heneicosyl, docosyl, tricosyl, tetracosyl, pentacosyl, hexacosyl, heptacosyl, octacosyl, nonacosyl, triacontyl, hentriacontyl, dotriacontyl, tritriacontyl, tetratriacontyl, pentatriacontyl, hexatriacontyl, heptatriacontyl, octatriacontyl, nonatriacontyl, tetracontyl, linoleyl((9Z,12Z)-octadeca-9,12-dien-1-yl), palmitoleyl((Z)-hexadec-9-en-1-yl), oleyl((Z)-octadec-9-en-1-yl), elaidyl(trans-9-octadecenyl), cis-vaccenyl(cis-11-octadecenyl), gadoleyl((Z)-icos-9-en-1-yl), 11-eicosenyl, erucyl(cis-13-docosenyl), 15-tetracosen-1-yl, eicosadienyl(icoso-11,14-dien-1-yl), linolenyl((9Z,12Z,15Z)-9,12,15-octadecatrien-1-yl), γ -linolenyl((6E,9E,12E)-octadeca-6,9,12-trien-1-yl), eleostearyl(octadeca-9,11,13-trien-1-yl), icoso-5,8,11-trien-1-yl, eicos-13-en-1-yl, icoso-11,14-17-trien-1-yl, octadeca-6,9,12,15-tetraen-1-yl, arachidonyl((5Z,8Z,11Z,14Z)-icosatetraen-1-yl), 4E,6Z-hexadecadien-1-yl, icoso-5,8,11,14,17-pentaen-1-yl, docosahexaenoyl(docosa-4,7,10,13,16,19-hexaen-1-yl), docosa-7,10,13,16,19-pentaen-1-yl, and tetracosa-6,9,12,15,18,21-hexaen-1-yl. In some embodiments, R.sup.b is linoleyl.

[0150] In some embodiments, the L has a formula (D), (E), or (F):

##STR00019## [0151] wherein: [0152] n, p, and q are each independently 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, or 40; and [0153] R.sup.a1 and R.sup.a2 are each independently selected from C.sub.6-C.sub.40 alkyl and C.sub.6-C.sub.40 alkenyl.

[0154] In some embodiments, L has formula (D). In some embodiments, L has formula (E). In some embodiments, L has formula (F). In some embodiments, n is 1, 2, 3, 4, 5, 6, 7, or 8. In some embodiments, n is 2, 3, or 4. In some embodiments, n is 2. In some embodiments, n is 3. In some

embodiments, n is 4.

[0155] In some embodiments, p and q are each independently selected from 1, 2, 3, 4, 5, 6, 7, or 8. In some embodiments, p and q are each independently selected from 3, 4, 5, 6, and 7. In some embodiments, p and q are each independently selected from 3, 5, and 7. In some embodiments, p and q are each 5. In some embodiments, p and q are each 6. In some embodiments, p and q are each 7.

[0156] In some embodiments, R_{sup.a1} and R_{sup.a2} are each independently selected from C_{sub.6}-C_{sub.24} alkyl and C_{sub.6}-C_{sub.24} alkenyl. In some embodiments, R_{sup.a1} and R_{sup.a2} are each independently selected from C_{sub.9}-C_{sub.22} alkyl and C_{sub.9}-C_{sub.22} alkenyl. In some embodiments, R_{sup.a1} and R_{sup.a2} are each independently selected from straight or branched C_{sub.6}, C_{sub.7}, C_{sub.8}, C_{sub.9}, C_{sub.10}, C_{sub.11}, C_{sub.12}, C_{sub.13}, C_{sub.14}, C_{sub.15}, C_{sub.16}, C_{sub.17}, C_{sub.18}, C_{sub.19}, C_{sub.20}, C_{sub.21}, C_{sub.22}, C_{sub.23}, or C_{sub.24} alkyl. In some embodiments, R_{sup.a1} and R_{sup.a2} are each independently selected from straight or branched C_{sub.6}, C_{sub.7}, C_{sub.8}, C_{sub.9}, C_{sub.10}, C_{sub.11}, C_{sub.12}, C_{sub.13}, C_{sub.14}, C_{sub.15}, C_{sub.16}, C_{sub.17}, C_{sub.18}, C_{sub.19}, C_{sub.20}, C_{sub.21}, C_{sub.22}, C_{sub.23}, or C_{sub.24} alkenyl. In some embodiments, R_{sup.a1} and R_{sup.a2} are each independently selected from linoleyl, n-nonyl, n-undecyl, heneicosan-11-yl, pentadecane-7-yl, and heptadecan-9-yl.

[0157] In some embodiments, L is derived from a steroid. For example, in some embodiments, L is derived from cholesterol, beta-sitosterol, or BHEM-cholesterol.

[0158] In some embodiments, L is selected from:

##STR00020## ##STR00021## ##STR00022##

[0159] The compositions may take the form of a liposome, a lipid nanoparticle, a micelle, or the like. Methods of making lipid compositions include, for example, lipid film hydration, optionally coupled with sonication or extrusion, solvent evaporation (e.g., ethanol injection, ether injection, or reverse phase evaporation), solvent-diffusion method, hot homogenization process, detergent removal methods, or combinations thereof. Any naturally occurring or synthetic vesicle forming lipid or combinations thereof can be used. The one or more vesicle forming lipids may be selected from di-aliphatic chain lipids, such as phospholipids; diglycerides; di-aliphatic glycolipids; single lipids such as sphingomyelin or glycosphingolipid; steroidal lipids; hydrophilic polymer derivatized lipids; or mixtures thereof.

[0160] Lipid compositions of the disclosure may include one or more cationic and/or ionizable lipids, phospholipids, neutral or non-cationic lipids, polyethylene glycol (PEG)-lipid conjugates, and/or structural lipids.

[0161] Cationic and/or ionizable lipids include, for example, amine-containing lipids that can be readily protonated and may have a positive or partial positive charge at physiological pH due to a pK_a value between pH 5 and 8. The polar headgroup of the cationic lipids preferably comprises amine derivatives such as primary, secondary, and/or tertiary amines, quaternary ammonium, various combinations of amines, amidinium salts, or guanidine and/or imidazole groups as well as pyridinium, piperazine and amino acid headgroups such as lysine, arginine, ornithine and/or tryptophan. Cationic lipids include, but are not limited to, 1,2-dimyristoyl-sn-glycero-3-ethylphosphocholine (DMEPC), 1,2-di-O-octadecenyl-3-trimethylammonium propane (DOTMA) and/or 1,2-dioleoyl-3-trimethylammonium propane (DOTAP), 1,2-dimyristoyl-3-trimethylammonium propane (DMTAP), 2,3-di(tetradecoxy)propyl-(2-hydroxyethyl)-dimethylazanium bromide (DMRIE), didodecyl(dimethyl)ammonium bromide (DDAB), 1,2-dioleoyloxypropyl-3-dimethyl-hydroxyethyl ammonium bromide (DORIE), 3β-[N-(N,N'-dimethylamino-ethane)carbamoyl]cholesterol (DC-Chol) or dioleyl ether phosphatidylcholine (DOEPC). Ionizable lipids include, but are not limited to, 1,2-dioleoyloxy-3-dimethylamino-propane (DODMA).

[0162] In some embodiments, the disclosed compounds are incorporated into lipid compositions

comprising one or more phospholipids. Phospholipids include a phospholipid moiety and one or more fatty acid moieties. A phospholipid moiety may include, but is not limited to, phosphatidyl choline, phosphatidyl ethanolamine, phosphatidyl glycerol, phosphatidyl serine, phosphatidic acid, 2-lysophosphatidyl choline, and sphingomyelin. A fatty acid moiety may include, but is not limited to, lauric acid, myristic acid, myristoleic acid, palmitic acid, palmitoleic acid, stearic acid, oleic acid, linoleic acid, alpha-linolenic acid, erucic acid, phytanic acid, arachidic acid, arachidonic acid, eicosapentaenoic acid, behenic acid, docosapentaenoic acid, and docosahexaenoic acid.

[0163] Phospholipids suitable for use in the compositions may include, but are not limited to, phosphatidylglycerol (PG) including dimyristoyl phosphatidylglycerol (DMPG) and 1,2-dioleoyl-sn-glycero-3-phospho-rac-(1-glycerol) sodium salt (DOPG); phosphatidylcholine (PC), including egg yolk phosphatidylcholine, dimyristoyl phosphatidylcholine (DMPC), 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC), 1,2-dilinoleoyl-sn-glycero-3-phosphocholine (DLPC), 1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC), 1,2-dipalmitoyl-sn-glycero-3-phosphocholine (DPPC), 1,2-diundecanoyl-sn-glycero-phosphocholine (DUPC), 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine (POPC), 1,2-di-O-octadecenyl-sn-glycero-3-phosphocholine (18:0 Diether PC), 1-oleoyl-2-cholesterylhemisuccinoyl-sn-glycero-3-phosphocholine (OChemPC), 1-hexadecyl-sn-glycero-3-phosphocholine (C16 Lyso PC), 1,2-dilinolenoyl-sn-glycero-3-phosphocholine, 1,2-didocosahexaenoyl-sn-glycero-3-phosphocholine, 1,2-diarachidonoyl-sn-glycero-3-phosphocholine; phosphatidylethanolamine (PE) including 1,2-dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE), 1,2-diphytanoyl-sn-glycero-3-phosphoethanolamine (ME 16.0 PE), 1,2-distearoyl-sn-glycero-3-phosphoethanolamine, 1,2-dilinoleoyl-sn-glycero-3-phosphoethanolamine, 1,2-dilinolenoyl-sn-glycero-3-phosphoethanolamine, 1,2-diarachidonoyl-sn-glycero-3-phosphoethanolamine, 1,2-didocosahexaenoyl-sn-glycero-3-phosphoethanolamine; phosphatidic acid (PA); phosphatidylinositol (PI); phosphatidylserine (PS); and sphingomyelin (SM).

[0164] In some embodiments, the lipid compositions comprise a structural lipid. In some embodiments, the structural lipid is a sterol. The sterol may comprise cholesterol, fecosterol, ergosterol, campesterol, sitosterol, stigmasterol, brassicasterol, or a sterol ester, such as cholesteryl hemisuccinate, cholesteryl sulfate, ursolic acid, alpha-tocopherol, or any other derivatives of cholesterol.

[0165] In some embodiments, the lipid compositions comprise a polyethylene glycol (PEG)-lipid conjugate. A PEG-lipid conjugate may include, but is not limited to, PEG-modified phosphatidylethanolamines, PEG-modified phosphatidic acids, PEG-modified ceramides, PEG-modified dialkylamines, PEG-modified diacylglycerols, PEG-modified dialkylglycerols, and mixtures thereof. For example, a PEG lipid may be PEG-DMG (1,2-dimyristoyl-rac-glycero-3-methoxypolyethylene glycol), PEG-c-DOMG (R-3-[(ω -methoxy poly(ethylene glycol)2000)carbamoyl]-1,2-dimyristyloxylpropyl-3-amine), PEG-DMA (PEG-dimethacrylate), PEG-DLPE (1,2-didodecanoyl-sn-glycero-3-phosphoethanolamine-PEG), PEG-DMPE (PEG-1,2-dimyristoyl-sn-glycero-3-phosphoethanolamine), PEG-DPPC (PEG-dipalmitoyl phosphatidylcholine), PEG-N,N-di(tetradecyl)acetamide, or a PEG-DSPE (1,2-distearoyl-sn-glycero-3-phosphoethanolamine-poly(ethylene glycol)) lipid. In some embodiments, the lipid nanoparticle comprises PEG-DMG and/or PEG-N,N-di(tetradecyl) acetamide.

[0166] The lipid structures described herein may also include other components typically used in the formation of vesicles (e.g., for stabilization). Examples of such other components includes, without being limited thereto, fatty alcohols, fatty acids, and/or any other pharmaceutically acceptable excipients which may affect the surface charge, the membrane fluidity and assist in the incorporation of the lipid into the lipid assembly.

[0167] The lipid compositions can also be targeting, e.g., contain one or more targeting moieties or biodistribution modifiers on the surface. A targeting moiety can be any agent that is capable of specifically binding or interacting with a desired target and are generally known in the art, for

example ligands such as folic acid, proteins, antibody or antibody fragments, and the like). In some embodiments, the targeting moiety is an immune cell epitope (e.g., B cell and T cell epitopes). In select embodiments, the targeting moiety comprises one or more epitopes from a microbiological agent (e.g., *Clostridioides difficile*, *Bacillus anthracis*, *Clostridium botulinum*, *Helicobacter pylori*, *Rotavirus* sp., Coronaviridae).

[0168] The lipid compositions can have any structure, e.g., structures having an inner space sequestered from the outer medium by one or more lipid bilayers, or any microcapsule that has a semi-permeable membrane with a lipophilic central part where the membrane sequesters an interior. In some embodiments, the lipid compositions may comprise unilamellar liposomes, having a single lipid layer. In some embodiments, the lipid compositions may comprise liposomes having a lipid bilayer. In some embodiments, the lipid compositions comprise micelles.

4. Active Agents

[0169] Any of the compositions disclosed herein may further comprise at least one active agent. In some embodiments the at least one active agent comprises a nucleic acid (mRNA, aptamers, antisense oligonucleotides, ribozyme nucleic acids, interfering RNAs, antisense and antigenic nucleic acids), a protein, an immune modulator, a chemotherapeutic agent, a steroid, an analgesic, an antimicrobial agent, or a combination thereof. In some embodiments, the at least one active agent is encapsulated by a liposome, a lipid nanoparticle, a micelle in the lipid composition.

[0170] In some embodiments, the at least one active agent comprises a nucleic acid (e.g., ribonucleic acid or deoxyribonucleic acid). Exemplary nucleic acids for use in accordance with the present disclosure include, but are not limited to, one or more of deoxyribonucleic acid (DNA), ribonucleic acid (RNA) including messenger mRNA (mRNA), hybrids thereof, RNAi-inducing agents, RNAi agents, siRNAs, shRNAs, miRNAs, antisense RNAs, ribozymes, catalytic DNA, RNAs that induce triple helix formation, aptamers, vectors, etc. In some embodiments, the nucleic acid encodes an antigen or a functional fragment thereof.

[0171] In some embodiments, the at least one active agent is an RNA. RNAs useful in the compositions and methods described herein can be selected from the group consisting of, but are not limited to, shortmers, antagomirs, antisense RNAs, ribozymes, small interfering RNA (siRNA), asymmetrical interfering RNA (aiRNA), microRNA (miRNA), Dicer-substrate RNA (dsRNA), small hairpin RNA (shRNA), transfer RNA (tRNA), messenger RNA (mRNA), and mixtures thereof.

[0172] In certain embodiments, the at least one active agent is an mRNA. An mRNA may encode any polypeptide of interest, including any naturally or non-naturally occurring or otherwise modified polypeptide. A polypeptide encoded by an mRNA may be of any size and may have any secondary structure or activity. In some embodiments, a polypeptide encoded by an mRNA may have a therapeutic effect when expressed in a cell.

[0173] In other embodiments, the at least one active agent is an siRNA. An siRNA may be capable of selectively knocking down or down regulating expression of a gene of interest. For example, an siRNA could be selected to silence a gene associated with a particular disease, disorder, or condition upon administration to a subject in need thereof of a nanoparticle composition including the siRNA. An siRNA may comprise a sequence that is complementary to an mRNA sequence that encodes a gene or protein of interest. In some embodiments, the siRNA may be an immunomodulatory siRNA.

[0174] In some embodiments, the at least one active agent is an shRNA or a vector or plasmid encoding the same. An shRNA may be produced inside a target cell upon delivery of an appropriate construct to the nucleus. Constructs and mechanisms relating to shRNA are well known in the relevant arts.

[0175] In some embodiments, the at least one active agent is an immune modulator. Exemplary immune modulators include: signal transducer and activator of transcription 3 (Stat3) inhibitors and analogs thereof, such as, SM-36 and its analogs; toll-like receptor (TLR) agonists and analogs

thereof, such as, imiquimod, resiquimod, selgantolimod, gardiquimod, SM-360320, TMX-101, TMX-202, TMX-302, TMX-306, GSK2245035, CL097, 852A, AZD-8848, DSP-3025, GS-9620, RO7020531, RO6871765, ANA773, DSP-0509, NJH395, BNT411, TQ-A3334, JNJ-4964, LHC165, CV8102, VTX-1463, VTX-2337, IMO-8400, IMO-3100, IRS-954, and analogs thereof; and statins or other lipid-lowering medications and analogs thereof, such as, atorvastatin, pravastatin, fluvastatin, simvastatin, lovastatin, mevastatin, pitavastatin, rosuvastatin, and analogs thereof.

[0176] In some embodiments, the at least one active agent comprises at least one chemotherapeutic agent. As used herein, the term “chemotherapeutic” or “anti-cancer drug” includes any small molecule or other drug used in cancer treatment or prevention. Chemotherapeutics include, but are not limited to, cyclophosphamide, methotrexate, 5-fluorouracil, doxorubicin, docetaxel, daunorubicin, bleomycin, vinblastine, dacarbazine, cisplatin, paclitaxel, raloxifene hydrochloride, tamoxifen citrate, abemaciclib, afinitor (Everolimus), alpelisib, anastrozole, pamidronate, anastrozole, exemestane, capecitabine, epirubicin hydrochloride, eribulin mesylate, toremifene, fulvestrant, letrozole, gemcitabine, goserelin, ixabepilone, emtansine, lapatinib, olaparib, megestrol, neratinib, palbociclib, ribociclib, talazoparib, thiotepa, toremifene, methotrexate, and tucatinib.

[0177] In some embodiments, the at least one active agent comprises an antimicrobial (e.g., antiviral or antibacterial) agent. In some embodiments, the antimicrobial agent is an antiviral agent, including but not limited to, abacavir, acyclovir, adefovir, amantadine, amprenavir, atazanavir, baloxavir marboxil, bictegravir, boceprevir, bulevirtide, cidofovir, cobicistat, daclatasvir, darunavir, delavirdine, didanosine, docosanol, dolutegravir, doravirine, edoxudine, efavirenz, elvitegravir, emtricitabine, enfuvirtide, entecavir, etravirine, famciclovir, fomivirsen, fosamprenavir, foscarnet, ganciclovir, ibacitabine, ibalizumab, idoxuridine, imiquimod, immunovir, indinavir, lamivudine, letermovir, lopinavir, loviride, maraviroc, methisazone, moroxydine, nelfinavir, nevirapine, nexavir, nitazoxanide, oseltamivir, penciclovir, peramivir, penciclovir, peramivir, pleconaril, podophyllotoxin, raltegravir, remdesivir, ribavirin, rilpivirine, rilpivirine, rimantadine, rintatolimod, ritonavir, saquinavir, simeprevir, sofosbuvir, stavudine, taribavirin, telaprevir, telbivudine, tenofovir (e.g., tenofovir alafenamide or tenofovir disoproxil), tipranavir, trifluridine, tromantadine, umifenovir, valaciclovir, valganciclovir, vicriviroc, vidarabine, zalcitabine, zanamivir, and zidovudine, and any combination thereof.

[0178] In some embodiments, the antimicrobial agent is an antibacterial agent. Exemplary antibacterial agents include sulfonamides, amphenicols, spectinomycin, trimethoprim, glycylicyclines, macrolides (e.g., erythromycin, clarithromycin, azithromycin, roxithromycin), oxazolidinones (e.g., linezolid), tetracyclines (e.g., doxycycline, tetracycline, minocycline), β -lactams (e.g., penicillin, methicillin, cloxacillin), carbapenems (e.g., imipenem, meropenem, aztreonam), aminoglycosides (e.g., gentamicin, tobramycin, amikacin), quinolones and fluoroquinolones (e.g., levofloxacin, ciprofloxacin, moxifloxacin), glycopeptides (e.g., vancomycin), polymyxins (e.g., polymyxin, colistin).

5. Pharmaceutical Compositions

[0179] The lipid compositions may be incorporated into pharmaceutically acceptable compositions. The pharmaceutical compositions and formulations may include pharmaceutically acceptable carriers. The term “pharmaceutically acceptable carrier,” as used herein, means a non-toxic, inert solid, semi-solid or liquid filler, diluent, encapsulating material, surfactant, cyclodextrins or formulation auxiliary of any type. Some examples of materials which can serve as pharmaceutically acceptable carriers are sugars such as, but not limited to, lactose, glucose and sucrose; starches such as, but not limited to, corn starch and potato starch; cellulose and its derivatives such as, but not limited to, sodium carboxymethyl cellulose, ethyl cellulose and cellulose acetate; powdered tragacanth; malt; gelatin; talc; excipients such as, but not limited to, cocoa butter and suppository waxes; oils such as, but not limited to, peanut oil, cottonseed oil,

safflower oil, sesame oil, olive oil, corn oil and soybean oil; surfactants such as, but not limited to, cremophor EL, cremophor RH 60, Solutol HS 15 and polysorbate 80; cyclodextrins such as, but not limited to, alpha-CD, beta-CD, gamma-CD, HP-beta-CD, SBE-beta-CD; glycols; such as propylene glycol; esters such as, but not limited to, ethyl oleate and ethyl laurate; agar; buffering agents such as, but not limited to, magnesium hydroxide and aluminum hydroxide; alginic acid; pyrogen-free water; isotonic saline; Ringer's solution; ethyl alcohol, and phosphate buffer solutions, as well as other non-toxic compatible lubricants such as, but not limited to, sodium lauryl sulfate and magnesium stearate, as well as coloring agents, releasing agents, coating agents, sweetening, flavoring and perfuming agents, preservatives and antioxidants can also be present in the composition, according to the judgment of the formulator.

[0180] The route by which the compositions are administered and the form of the composition will dictate the type of carrier to be used. The composition may be in a variety of forms, suitable, for example, for systemic administration (e.g., oral, rectal, nasal, sublingual, buccal, implants, or parenteral injections) or topical administration (e.g., dermal, pulmonary, nasal, aural, ocular, liposome delivery systems, or iontophoresis).

[0181] Carriers for systemic administration typically include at least one of diluents, lubricants, binders, disintegrants, colorants, flavors, sweeteners, antioxidants, preservatives, glidants, solvents, suspending agents, wetting agents, surfactants, cyclodextrins combinations thereof, and others. All carriers are optional in the compositions.

[0182] Suitable diluents include sugars such as glucose, lactose, dextrose, and sucrose; diols such as propylene glycol; calcium carbonate; sodium carbonate; sugar alcohols, such as glycerin; mannitol; and sorbitol. The amount of diluent(s) in a systemic or topical composition is typically about 50 to about 90%.

[0183] Suitable lubricants include silica, talc, stearic acid and its magnesium salts and calcium salts, calcium sulfate; and liquid lubricants such as polyethylene glycol and vegetable oils such as peanut oil, cottonseed oil, sesame oil, olive oil, corn oil and oil of theobroma. The amount of lubricant(s) in a systemic or topical composition is typically about 5 to about 10%.

[0184] Suitable binders include polyvinyl pyrrolidone; magnesium aluminum silicate; starches such as corn starch and potato starch; gelatin; tragacanth; and cellulose and its derivatives, such as sodium carboxymethylcellulose, ethyl cellulose, methylcellulose, microcrystalline cellulose, and sodium carboxymethylcellulose. The amount of binder(s) in a systemic composition is typically about 5 to about 50%.

[0185] Suitable disintegrants include agar, alginic acid and the sodium salt thereof, effervescent mixtures, croscarmellose, crospovidone, sodium carboxymethyl starch, sodium starch glycolate, clays, and ion exchange resins. The amount of disintegrant(s) in a systemic or topical composition is typically about 0.1 to about 10%.

[0186] Suitable colorants include a colorant such as an FD&C dye. When used, the amount of colorant in a systemic or topical composition is typically about 0.005 to about 0.1%.

[0187] Suitable flavors include menthol, peppermint, and fruit flavors. The amount of flavor(s), when used, in a systemic or topical composition is typically about 0.1 to about 1.0%.

[0188] Suitable sweeteners include aspartame and saccharin. The amount of sweetener(s) in a systemic or topical composition is typically about 0.001 to about 1%.

[0189] Suitable antioxidants include butylated hydroxyanisole ("BHA"), butylated hydroxytoluene ("BHT"), and vitamin E. The amount of antioxidant(s) in a systemic or topical composition is typically about 0.1 to about 5%.

[0190] Suitable preservatives include benzalkonium chloride, methyl paraben and sodium benzoate. The amount of preservative(s) in a systemic or topical composition is typically about 0.01 to about 5%.

[0191] Suitable glidants include silicon dioxide. The amount of glidant(s) in a systemic or topical composition is typically about 1 to about 5%.

[0192] Suitable solvents include water, isotonic saline, ethyl oleate, glycerine, hydroxylated castor oils, alcohols such as ethanol, dimethyl sulfoxide, N-Methyl-2-Pyrrolidone, dimethylacetamide and phosphate (or other suitable buffer). The amount of solvent(s) in a systemic or topical composition is typically from about 0 to about 100%.

[0193] Suitable suspending agents include AVICEL RC-591 (from FMC Corporation of Philadelphia, Pa.) and sodium alginate. The amount of suspending agent(s) in a systemic or topical composition is typically about 1 to about 8%.

[0194] Suitable surfactants include lecithin, Polysorbate 80, and sodium lauryl sulfate, and the TWEENS from Atlas Powder Company of Wilmington, Del. Suitable surfactants include those disclosed in the C.T.F.A. Cosmetic Ingredient Handbook, 1992, pp. 587-592; Remington's Pharmaceutical Sciences, 15th Ed. 1975, pp. 335-337; and McCutcheon's Volume 1, Emulsifiers & Detergents, 1994, North American Edition, pp. 236-239. The amount of surfactant(s) in the systemic or topical composition is typically about 0.1% to about 5%.

[0195] Suitable cyclodextrins include alpha-CD, beta-CD, gamma-CD, hydroxypropyl betadex (HP-beta-CD), sulfobutyl-ether β -cyclodextrin (SBB-beta-CD). The amount of cyclodextrins in the systemic or topical composition is typically about 0% to about 40%.

[0196] Compositions for oral administration can have liquid forms. For example, suitable liquid forms include aqueous solutions, emulsions, suspensions, solutions reconstituted from non-effervescent granules, suspensions reconstituted from non-effervescent granules, effervescent preparations reconstituted from effervescent granules, elixirs, tinctures, syrups, and the like. Liquid orally administered compositions typically include a disclosed compound and a carrier, namely, a carrier selected from diluents, colorants, flavors, sweeteners, preservatives, solvents, suspending agents, and surfactants. Peroral liquid compositions preferably include one or more ingredients selected from colorants, flavors, and sweeteners.

[0197] Other compositions useful for attaining systemic delivery of the subject compounds include sublingual, buccal and nasal dosage forms. Such compositions typically include one or more of soluble filler substances such as diluents including sucrose, sorbitol, and mannitol; and binders such as acacia, microcrystalline cellulose, carboxymethyl cellulose, and hydroxypropyl methylcellulose. Such compositions may further include lubricants, colorants, flavors, sweeteners, antioxidants, and glidants.

[0198] The amount of the carrier employed in conjunction with a disclosed composition is sufficient to provide a practical quantity of composition for administration per unit dose of the compound. Techniques and compositions for making dosage forms useful in the methods of this invention are described in the following references: Modern Pharmaceutics, Chapters 9 and 10, Banker & Rhodes, eds. (1979); Lieberman et al., Pharmaceutical Dosage Forms: Tablets (1981); and Ansel, Introduction to Pharmaceutical Dosage Forms, 2nd Ed., (1976).

[0199] A carrier may include a single ingredient or a combination of two or more ingredients.

6. Vaccines

[0200] The compositions may also be used for vaccines. The vaccines comprise the compositions disclosed and an antigen or a nucleic acid encoding thereof. Suitable antigens include microbial pathogens, bacteria, viruses, proteins, glycoproteins lipoproteins, peptides, glycopeptides, lipopeptides, toxoids, carbohydrates, and tumor-specific antigens. Mixtures of two or more antigens may be employed.

[0201] The antigen can be derived and/or isolated from essentially any desired source depending on the infectious disease, autoimmune disease, condition, cancer, pathogen, or a disease that is to be treated with a given vaccine composition.

[0202] The vaccines described herein may be capable of providing immunity against one or more conditions related to infectious diseases, including but not limited to, influenza, measles, human papillomavirus (HPV), rabies, meningitis, whooping cough, tetanus, plague, hepatitis, and tuberculosis and can include infectious disease derived antigens and/or epitopes, or nucleic acids

encoding thereof.

[0203] The vaccines described herein may also direct an immune response against cancer cells and can include tumor cell derived antigens, epitopes, and/or neoepitopes, or portions thereof, or nucleic acids encoding tumor cell derived antigens, epitopes, and/or neoepitopes. Tumor antigens are surface molecules that are differentially expressed in tumor cells relative to non-tumor tissues. Tumor antigens make tumor cells immunologically distinct from normal cells and provide diagnostic and therapeutic targets for human cancers. Tumor antigens have been characterized either as membrane proteins or as altered carbohydrate molecules of glycoproteins or glycolipids on the cell surface. Cancer cells often have distinctive tumor antigens on their surfaces, such as truncated epidermal growth factor, folate binding protein, epithelial mucins, melanoferrin, carcinoembryonic antigen, prostate-specific membrane antigen, HER2-neu, which are candidates for use in therapeutic cancer vaccines. Because tumor antigens are normal or related to normal components of the body, the immune system often fails to mount an effective immune response against those antigens to destroy the tumor cells. Illustrative cancer types for which this approach can be used include prostate, colon, breast, ovarian, pancreatic, brain, head and neck, melanoma, leukemia, lymphoma, etc.

[0204] In other embodiments, the antigen present in the vaccine composition is not a foreign antigen, but a self-antigen, e.g., the vaccine composition is directed toward an autoimmune disease. Examples of autoimmune diseases include type 1 diabetes, conventional organ specific autoimmunity, neurological disease, rheumatic diseases/connective tissue disease, autoimmune cytopenias, and related autoimmune diseases. Such conventional organ specific autoimmunity may include thyroiditis (Graves+Hashimoto's), gastritis, adrenalitis (Addison's), ovaritis, primary biliary cirrhosis, myasthenia gravis, gonadal failure, hypoparathyroidism, alopecia, malabsorption syndrome, pernicious anemia, hepatitis, anti-receptor antibody diseases and vitiligo. Such neurological diseases may include schizophrenia, Alzheimer's disease, depression, hypopituitarism, diabetes insipidus, sicca syndrome and multiple sclerosis. Such rheumatic diseases/connective tissue diseases may include rheumatoid arthritis, systemic lupus erythematosus (SLE) or Lupus, scleroderma, polymyositis, inflammatory bowel disease, dermatomyositis, ulcerative colitis, Crohn's disease, vasculitis, psoriatic arthritis, exfoliative psoriatic dermatitis, pemphigus vulgaris, Sjogren's syndrome. Other autoimmune related diseases may include autoimmune uveoretinitis, glomerulonephritis, post myocardial infarction cardiomyopathy syndrome, pulmonary hemosiderosis, amyloidosis, sarcoidosis, aphthous stomatitis, and other immune related diseases, as presented herein and known in the related arts.

[0205] In one embodiment, the antigen in a vaccine composition is a peptide, polypeptide, or immunogenic portion thereof. An "immunogenic portion," as used herein is a portion of a protein that is recognized (e.g., specifically bound) by a B cell and/or T cell surface antigen receptor. Such immunogenic portions generally comprise at least 5 amino acid residues, more preferably at least 10, and still more preferably at least 20 amino acid residues of an antigenic protein or a variant thereof.

[0206] Immunogenic portions of antigen polypeptides may generally be identified using well known techniques, such as those summarized in Paul, Fundamental Immunology, 3rd ed., 243-247 (Raven Press, 1993) and references cited therein. Such techniques include screening polypeptides for the ability to react with antigen-specific antibodies, antisera and/or T cell lines or clones. As used herein, antisera and antibodies are "antigen-specific" if they specifically bind to an antigen (e.g., they react with the protein in an ELISA or other immunoassay, and do not react detectably with unrelated proteins). Such antisera and antibodies may be prepared using known techniques. An immunogenic portion of a protein is a portion that reacts with such antisera and/or T cells at a level that is not substantially less than the reactivity of the full length polypeptide (e.g., in an ELISA and/or T cell reactivity assay). Such immunogenic portions may react within such assays at a level that is similar to or greater than the reactivity of the full length polypeptide. Such screens

may generally be performed using methods well known to those of ordinary skill in the art, such as those described in Harlow and Lane, *Antibodies: A Laboratory Manual*, Cold Spring Harbor Laboratory, 1988. For example, a polypeptide may be immobilized on a solid support and contacted with patient sera to allow binding of antibodies within the sera to the immobilized polypeptide. Unbound sera may then be removed and bound antibodies detected using, for example, ¹²⁵I-labeled Protein A.

[0207] Peptide and polypeptide antigens may be prepared using any of a variety of well-known techniques. Recombinant polypeptides encoded by DNA sequences may be readily prepared from isolated DNA sequences using any of a variety of expression vectors known to those of ordinary skill in the art. Expression may be achieved in any appropriate host cell that has been transformed or transfected with an expression vector containing a DNA molecule that encodes a recombinant polypeptide. Suitable host cells include prokaryotes, yeast, and higher eukaryotic cells, such as mammalian cells and plant cells. Preferably, the host cells employed are *E. coli*, yeast, or a mammalian cell line such as COS or CHO.

[0208] Portions and other variants of a protein antigen having less than about 100 amino acids, and generally less than about 50 amino acids, may also be generated by synthetic means, using techniques well known to those of ordinary skill in the art. For example, such polypeptides may be synthesized using any of the commercially available solid-phase techniques, such as the Merrifield solid-phase synthesis method, where amino acids are sequentially added to a growing amino acid chain. See, Merrifield, *J. Am. Chem. Soc.* 85:2149-2146, 1963. Equipment for automated synthesis of polypeptides is commercially available from suppliers such as Perkin Elmer/Applied BioSystems Division (Foster City, Calif.), and may be operated according to the manufacturer's instructions.

[0209] In certain embodiments, the nucleic acid encoding the antigen is DNA. Illustrative DNA-based vaccines of this type contain DNA encoding one or more polypeptide antigens, such that the antigen is generated in situ. Alternatively, the vaccine may be an RNA-based vaccine. In certain embodiments, the nucleic acid encoding the antigen is an mRNA. An mRNA may encode any polypeptide antigen of interest, including any naturally or non-naturally occurring or otherwise modified polypeptide. A polypeptide encoded by an mRNA may be of any size and may have any secondary structure or activity. In some embodiments, a polypeptide encoded by the mRNA may stimulate an immune response when expressed in a cell.

[0210] The vaccine compositions of the present disclosure may also contain other compounds, which may be biologically active or inactive. The vaccine or medicament may comprise an adjuvant or immunostimulant, or a polynucleotide encoding an adjuvant or immunostimulant (e.g., an adjuvative polypeptide). Adjuvants and immunostimulants are compounds or compositions that either directly or indirectly stimulate the immune system's response to a co-administered antigen. In some embodiments, the vaccines are not adjuvanted or are self-adjuvanting.

[0211] Suitable adjuvants are commercially available as, for example, Glucopyranosyl Lipid Adjuvant (GLA); Pam3CSK4; Freund's Incomplete Adjuvant and Complete Adjuvant (Difco Laboratories, Detroit, Mich.); Merck Adjuvant 65 (Merck and Company, Inc., Rahway, N.J.); AS-2 (SmithKline Beecham); mineral salts (for example, aluminum, silica, kaolin, and carbon); aluminum salts such as aluminum hydroxide gel (alum), AlK(SO₄)₂, AlNa(SO₄)₂, AlNH₄(SO₄), and Al(OH)₃; salts of calcium (e.g., Ca₃(PO₄)₂), iron or zinc; an insoluble suspension of acylated tyrosine; acylated sugars; cationically or anionically derivatized polysaccharides; polynucleotides (for example, poly IC, poly AU acids, and CpG oligodeoxynucleotides (e.g., Class A or B)); polyphosphazenes; cyanoacrylates; polymerase-(DL-lactide-co-glycoside); bovine serum albumin; diphtheria toxoid; tetanus toxoid; edestin; keyhole-limpet hemocyanin; Pseudomonas Toxin A; cholera toxin; pertussis toxin; viral proteins; Quil A; aminoalkyl glucosamine phosphate compounds. In addition, adjuvants such as cytokines (e.g., GM-CSF or interleukin-2, -7, or -12), interferons, or

tumor necrosis factor, may also be used as adjuvants. Protein and polypeptide adjuvants may be obtained from natural or recombinant sources according to methods well known to those skilled in the art. When obtained from recombinant sources, the adjuvant may comprise a protein fragment comprising at least the immunostimulatory portion of the molecule.

[0212] Other known immunostimulatory macromolecules which can be used include, but are not limited to, polysaccharides, tRNA, non-metabolizable synthetic polymers such as polyvinylamine, polymethacrylic acid, polyvinylpyrrolidone, mixed polycondensates (with relatively high molecular weight) of 4',4'-diaminodiphenylmethane-3,3'-dicarboxylic acid and 4-nitro-2-aminobenzoic acid (See, Sela, M., *Science* 166:1365-1374 (1969)) or glycolipids, lipids or carbohydrates.

[0213] In some embodiments, the adjuvantive polypeptide comprises immune activator proteins, such as CD70, CD40 ligand, and constitutively active TLR4, or polycationic peptides (e.g., protamine). In some embodiments, the adjuvantive polypeptide is a flagellin polypeptide. Commercially available mRNA encoding adjuvantive polypeptides are available, for example, as TriMix (See Bonehill, A. et al. *Mol. Ther.* 16, 1170-1180 (2008), incorporated herein by reference). In some embodiments, the vaccine may comprise at least two separate polynucleotides, one encoding anti-Müllerian hormone receptor II extracellular domain (AMHR2-ED), as described above, and the other encoding an adjuvantive polypeptide (e.g., a flagellin polypeptide or immune activator protein).

[0214] Vaccine preparation is a well-developed art and general guidance in the preparation and formulation of vaccines is readily available from any of a variety of sources. One such example is *New Trends and Developments in Vaccines*, edited by Volier et al. University Park Press, Baltimore, Md., U.S.A. 1978. Vaccine compositions may generally be used for prophylactic and therapeutic purposes.

[0215] The amount of antigen in each vaccine dose is generally selected as an amount which induces an immunoprotective response without significant adverse side effects in typical vaccines. Such amount will vary depending upon which specific immunogen is employed and how it is presented. Of course, the dosage administered may be dependent upon the age, weight, kind of concurrent treatment, if any, and nature of the antigen administered.

[0216] The immunogenic activity of a given amount of a vaccine composition can be readily determined, for example by monitoring the increase in titer of antibody against the antigen used in the vaccine composition (Dalsgaard, K. *Acta Veterinaria Scandinavica* 69:1-40 (1978)). Another common method involves injecting CD-1 mice intradermally with various amounts of a vaccine composition, later harvesting sera from the mice and testing for anti-immunogen antibody, e.g., by ELISA. These and other similar approaches will be apparent to the skilled artisan.

7. Methods of Use

[0217] The disclosure provides methods for delivering an active agent to a cell. The methods comprise contacting the cell with a composition as disclosed herein. In some embodiments, the cell is in vitro. In some embodiments, the cell is in vivo. Thus, the disclosure further provides methods for delivery any active agent to a subject. The methods may comprise administering a composition as described herein to a subject.

[0218] The disclosure also provides methods of producing a polypeptide of interest in a cell. The methods comprise contacting the cell with a composition as disclosed herein, wherein the composition comprises a mRNA encoding the polypeptide of interest. In some embodiments, the cell is in vitro. In some embodiments, the cell is in vivo. Thus, the disclosure provides methods of producing a polypeptide of interest in a subject comprise administering a composition as described herein to a subject, wherein the composition comprises a mRNA encoding the polypeptide of interest.

[0219] The disclosure further provides methods for treating a disease or disorder comprising administration of a composition as disclosed herein, to a subject in need thereof. In some embodiments, the subject is a human.

[0220] The disease or disorder may comprise cancer, autoimmune diseases, inflammatory diseases, and infectious diseases.

[0221] In some embodiments, the disease or disorder is an inflammatory disease or disorder. Inflammatory diseases are characterized by activation of the immune system in a tissue or an organ to abnormal levels that may lead to abnormal function and/or disease in the tissue or organ. The inflammatory diseases and disorders that may be treated by the methods of the present invention include, but are not limited to, arthritis, rheumatoid arthritis, asthma, inflammatory bowel disease (Crohn's disease or ulcerative colitis), chronic obstructive pulmonary disease (COPD), allergic rhinitis, vasculitis (polyarteritis nodosa, temporal arteritis, Wegener's granulomatosis, Takayasu's arteritis, or Behcet's syndrome), inflammatory neuropathy, psoriasis, systemic lupus erythematosus (SLE), chronic thyroiditis, Hashimoto's thyroiditis, Addison's disease, polymyalgia rheumatica, Sjogren's syndrome, or Churg-Strauss syndrome.

[0222] In some embodiments, the disease or disorder is an autoimmune disease or disorder. Autoimmune diseases and disorders refer to conditions in a subject characterized by cellular, tissue and/or organ injury caused by an immunologic reaction of the subject to its own cells, tissues and/or organs. Autoimmune diseases and disorders that may be treated by the methods of the present invention include, but are not limited to, alopecia areata, ankylosing spondylitis, antiphospholipid syndrome, autoimmune Addison's disease, autoimmune diseases of the adrenal gland, autoimmune hemolytic anemia, autoimmune hepatitis, autoimmune oophoritis and orchitis, autoimmune thrombocytopenia, Behcet's disease, bullous pemphigoid, cardiomyopathy, celiac sprue-dermatitis, chronic fatigue immune dysfunction syndrome (CFIDS), chronic inflammatory demyelinating polyneuropathy, Churg-Strauss syndrome, cicatricial pemphigoid, CREST syndrome, cold agglutinin disease, Crohn's disease, discoid lupus, essential mixed cryoglobulinemia, fibromyalgia-fibromyositis, glomerulonephritis, Graves' disease, Guillain-Barre, Hashimoto's thyroiditis, idiopathic pulmonary fibrosis, idiopathic thrombocytopenia purpura (ITP), irritable bowel disease (IBD), IgA neuropathy, juvenile arthritis, lichen planus, lupus erythematosus, Meniere's disease, mixed connective tissue disease, multiple sclerosis, type 1 or immune-mediated diabetes mellitus, myasthenia gravis, pemphigus vulgaris, pernicious anemia, polyarteritis nodosa, polychondritis, polyglandular syndromes, polymyalgia rheumatica, polymyositis and dermatomyositis, primary agammaglobulinemia, primary biliary cirrhosis, psoriasis, psoriatic arthritis, Raynaud's phenomenon, Reiter's syndrome, Rheumatoid arthritis, sarcoidosis, scleroderma, Sjogren's syndrome, stiff-man syndrome, systemic lupus erythematosus, lupus erythematosus, takayasu arteritis, temporal arteritis/giant cell arteritis, ulcerative colitis, uveitis, vasculitides such as dermatitis herpetiformis vasculitis, vitiligo, and Wegener's granulomatosis.

[0223] Some autoimmune disorders are also associated with an inflammatory condition. Examples of inflammatory disorders which are also autoimmune disorders that can be prevented, treated or managed in accordance with the methods of the invention include, but are not limited to, asthma, encephalitis, inflammatory bowel disease, chronic obstructive pulmonary disease (COPD), allergic disorders, pulmonary fibrosis, undifferentiated spondyloarthropathy, undifferentiated arthropathy, arthritis, inflammatory osteolysis, and chronic inflammation resulting from chronic viral or bacterial infections. Examples of the types of psoriasis which can be treated in accordance with the compositions and methods of the invention include, but are not limited to, plaque psoriasis, pustular psoriasis, erythrodermic psoriasis, guttate psoriasis and inverse psoriasis.

[0224] In some embodiments, the disease or disorder is cancer. In some embodiments, the cancer comprises a solid tumor. In some embodiments, the cancer comprises a blood cancer or lymphoma. In some embodiments, the cancer is metastatic cancer. In some embodiments, the disclosed compounds, compositions, or methods result in suppression of elimination of metastasis. In some embodiments, the disclosed compounds, compositions, or methods result in decreased tumor growth. In some embodiments, the disclosed compounds, compositions, or methods prevent tumor

recurrence.

[0225] The compounds and compositions herein may be useful to treat a wide variety of cancers including carcinoma, sarcoma, lymphoma, leukemia, melanoma, mesothelioma, multiple myeloma, or seminoma. The cancer may be a cancer of the bladder, blood, bone, brain, breast, cervix, colon/rectum, endometrium, head and neck, kidney, liver, lung, lymph nodes, muscle tissue, ovary, pancreas, prostate, skin, spleen, stomach, testicle, thyroid, or uterus.

[0226] In some embodiments, the cancer is invasive and/or metastatic cancer (e.g., stage II cancer, stage III cancer or stage IV cancer). In some embodiments, the cancer is an early stage cancer (e.g., stage 0 cancer, stage I cancer), and/or is not invasive and/or metastatic cancer.

[0227] In some embodiments, the disease or disorder is an infectious disease. Infectious diseases that can be treated or prevented by the methods of the present invention are caused by infectious agents including, but not limited to, viruses, bacteria, fungi, protozoa, helminths, and parasites. The invention is not limited to treating or preventing infectious diseases caused by intracellular or extracellular pathogens. The infectious disease may be derived from: bacteria, such as *Mycobacterium tuberculosis*, *Chlamydia*, *Francisella tularensis*; DNA viruses, such as Herpesviridae (herpes simplex virus-1, Kaposi's sarcoma-associated virus and Epstein-Barr virus), Papillomaviridae (human papilloma virus), Adenovirus and Hepadnaviridae (Hepatitis B virus), or RNA viruses, such as Retroviridae (human immunodeficiency virus) Flaviviridae (Dengue virus, Hepatitis C virus), Orthomyxoviridae (influenza), and Coronaviridae (human coronavirus and SARS coronavirus).

[0228] The compositions disclosed herein may be administered to a subject by a variety of methods. In any of the uses or methods described herein, administration may be by various routes known to those skilled in the art, including without limitation oral, inhalation, intravenous, intramuscular, topical, subcutaneous, systemic, and/or intraperitoneal administration to a subject in need thereof.

[0229] The amount of the compositions of the present disclosure required for use in the methods described herein will vary not only with the particular compound selected but also with the route of administration, the nature and/or symptoms of the disease and the age and condition of the patient and will be ultimately at the discretion of the attendant physician or clinician. The determination of effective dosage levels, that is the dosage levels necessary to achieve the desired result, can be accomplished by one skilled in the art using routine methods, for example, human clinical trials, in vivo studies, and in vitro studies. For example, useful dosages can be determined by comparing their in vitro activity, and in vivo activity in animal models.

[0230] Dosage amount and interval may be adjusted individually to provide plasma levels of the active moiety which are sufficient to maintain the modulating effects, or minimal effective concentration (MEC). The MEC will vary for each compound but can be estimated from in vivo and/or in vitro data. Dosages necessary to achieve the MEC will depend on individual characteristics and route of administration. However, FIPLC assays or bioassays can be used to determine plasma concentrations. Dosage intervals can also be determined using MEC value. Compositions should be administered using a regimen, which maintains plasma levels above the MEC for 10-90% of the time, preferably between 30-90% and most preferably between 50-90%. In cases of local administration or selective uptake, the effective local concentration of the drug may not be related to plasma concentration.

[0231] It should be noted that the attending physician would know how to and when to terminate, interrupt, or adjust administration due to toxicity or organ dysfunctions. Conversely, the attending physician would also know to adjust treatment to higher levels if the clinical response were not adequate (precluding toxicity). The magnitude of an administered dose in the management of the disorder of interest will vary with the severity of the symptoms to be treated and the route of administration. Further, the dose, and perhaps dose frequency, will also vary according to the age, body weight, and response of the individual patient. A program comparable to that discussed above

may be used in veterinary medicine.

[0232] The compositions disclosed herein can be evaluated for efficacy and toxicity using known methods. For example, the toxicology of a particular compound or a subset of the compounds sharing certain chemical moieties, or a composition thereof, may be established by determining in vitro toxicity towards a cell line, such as a mammalian, and preferably human, cell line. The results of such studies are often predictive of toxicity in animals, such as mammals, or more specifically, humans. Alternatively, the toxicity of particular compounds in an animal model, such as mice, rats, rabbits, dogs, or monkeys, may be determined using known methods. Efficacy may be established using several recognized methods, such as in vitro methods, animal models, or human clinical trials. When selecting a model to determine efficacy, the skilled artisan can be guided by the state of the art to choose an appropriate model, dose, route of administration and/or regime.

[0233] A wide range of second therapies may be used in conjunction with the compounds and methods of the present disclosure. The second therapy may be administration of an additional active agent or may be a second therapy not connected to administration of another agent. Such second therapies include, but are not limited to, surgery, immunotherapy, radiotherapy.

[0234] The second therapy may be administered at the same time as the initial therapy, either in the same composition or in a separate composition administered at substantially the same time as the first composition. In some embodiments, the second therapy may precede or follow the treatment of the first therapy by time intervals ranging from hours to months.

[0235] In some embodiments, the additional active agent comprises an immune modulator, a chemotherapeutic agent, a nucleic acid (e.g., mRNA, aptamers, antisense oligonucleotides, ribozyme nucleic acids, interfering RNAs, antigene nucleic acids), a steroid, an analgesic, and an antimicrobial agent, or a combination thereof.

[0236] Exemplary immune modulators include: signal transducer and activator of transcription 3 (Stat3) inhibitors and analogs thereof, such as, SM-36 and its analogs; toll-like receptor (TLR) agonists and analogs thereof, such as, imiquimod, resiquimod, selgantolimod, gardiquimod, SM-360320, TMX-101, TMX-202, TMX-302, TMX-306, GSK2245035, CL097, 852A, AZD-8848, DSP-3025, GS-9620, RO7020531, RO6871765, ANA773, DSP-0509, NJH395, BNT411, TQ-A3334, JNJ-4964, LHC165, CV8102, VTX-1463, VTX-2337, IMO-8400, IMO-3100, IRS-954, and analogs thereof; and statins or other lipid-lowering medications and analogs thereof, such as, atorvastatin, pravastatin, fluvastatin, simvastatin, lovastatin, mevastatin, pitavastatin, rosuvastatin, and analogs thereof.

[0237] In some embodiments, the additional active agent comprises at least one chemotherapeutic agent. As used herein, the term “chemotherapeutic” or “anti-cancer drug” includes any small molecule or other drug used in cancer treatment or prevention. Chemotherapeutics include, but are not limited to, cyclophosphamide, methotrexate, 5-fluorouracil, doxorubicin, docetaxel, daunorubicin, bleomycin, vinblastine, dacarbazine, cisplatin, paclitaxel, raloxifene hydrochloride, tamoxifen citrate, abemaciclib, afinitor (Everolimus), alpelisib, anastrozole, pamidronate, anastrozole, exemestane, capecitabine, epirubicin hydrochloride, eribulin mesylate, toremifene, fulvestrant, letrozole, gemcitabine, goserelin, ixabepilone, emtansine, lapatinib, olaparib, megestrol, neratinib, palbociclib, ribociclib, talazoparib, thiotepe, toremifene, methotrexate, and tucatinib. In select embodiments, the chemotherapeutic agent comprises paclitaxel.

[0238] In some embodiments, the compositions can be co-administered with an antimicrobial (e.g., antiviral or antibacterial) agent. In some embodiments, the antimicrobial agent is an antiviral agent, including but not limited to, abacavir, acyclovir, adefovir, amantadine, amprenavir, atazanavir, baloxavir marboxil, bictegravir, boceprevir, bulevirtide, cidofovir, cobicistat, daclatasvir, darunavir, delavirdine, didanosine, docosanol, dolutegravir, doravirine, edoxudine, efavirenz, elvitegravir, emtricitabine, enfuvirtide, entecavir, etravirine, famciclovir, fomivirsen, fosamprenavir, foscarnet, ganciclovir, ibacitabine, ibalizumab, idoxuridine, imiquimod, imunovir, indinavir, lamivudine, letermovir, lopinavir, loviride, maraviroc, methisazone, moroxydine, nelfinavir, nevirapine,

nexavir, nitazoxanide, oseltamivir, penciclovir, peramivir, pleconaril, podophyllotoxin, raltegravir, remdesivir, ribavirin, rilpivirine, rimantadine, rintatolimod, ritonavir, saquinavir, simeprevir, sofosbuvir, stavudine, taribavirin, telaprevir, telbivudine, tenofovir (e.g., tenofovir alafenamide or tenofovir disoproxil), tipranavir, trifluridine, tromantadine, umifenovir, valaciclovir, valganciclovir, vicriviroc, vidarabine, zalcitabine, zanamivir, and zidovudine, and any combination thereof.

[0239] In some embodiments, the antimicrobial agent is an antibacterial agent. Exemplary antibacterial agents include sulfonamides, amphenicols, spectinomycin, trimethoprim, glycylicyclines, macrolides (e.g., erythromycin, clarithromycin, azithromycin, roxithromycin), oxazolidinones (e.g., linezolid), tetracyclines (e.g., doxycycline, tetracycline, minocycline), β -lactams (e.g., penicillin, methicillin, cloxacillin), carbapenems (e.g., imipenem, meropenem, aztreonam), aminoglycosides (e.g., gentamicin, tobramycin, amikacin), quinolones and fluoroquinolones (e.g., levofloxacin, ciprofloxacin, moxifloxacin), glycopeptides (e.g., vancomycin), polymyxins (e.g., polymyxin, colistin).

[0240] In some embodiments, the second therapy includes immunotherapy. Immunotherapies include chimeric antigen receptor (CAR) T-cell or T-cell transfer therapies, cytokine therapy, immunomodulators, cancer vaccines, or administration of antibodies (e.g., monoclonal antibodies).

[0241] In some embodiments, the immunotherapy comprises administration of antibodies. The antibodies may target antigens either specifically expressed by tumor cells or antigens shared with normal cells. In some embodiments, the immunotherapy may comprise an antibody targeting, for example, CD20, CD33, CD52, CD30, HER (also referred to as erbB or EGFR), VEGF, CTLA-4 (also referred to as CD152), epithelial cell adhesion molecule (EpCAM, also referred to as CD326), and PD-1/PD-L1. Suitable antibodies include, but are not limited to, rituximab, blinatumomab, trastuzumab, gemtuzumab, alemtuzumab, ibritumomab, tositumomab, bevacizumab, cetuximab, panitumumab, ofatumumab, ipilimumab, brentuximab, pertuzumab, and the like). In some embodiments, the additional therapeutic agent may comprise anti-PD-1/PD-L1 antibodies, including, but not limited to, pembrolizumab, nivolumab, cemiplimab, atezolizumab, avelumab, durvalumab, and ipilimumab. The antibodies may also be linked to a chemotherapeutic agent. Thus, in some embodiments, the antibody is an antibody-drug conjugate.

[0242] The immunotherapy (e.g., administration of antibodies) may be administered to a subject by a variety of methods. In any of the uses or methods described herein, administration may be by various routes known to those skilled in the art, including without limitation oral, inhalation, intravenous, intramuscular, topical, subcutaneous, systemic, and/or intraperitoneal administration to a subject in need thereof. The immunotherapy may be administered by parenteral administration (including, but not limited to, subcutaneous, intramuscular, intravenous, intraperitoneal, intracardiac and intraarticular injections). In some embodiments, the immunotherapy may be administered in the same or different manner than the disclosed compounds or compositions.

8. Kits

[0243] In another aspect, the disclosure provides kits comprising at least one disclosed compound or a pharmaceutically acceptable salt thereof, or a composition comprising the compound or a pharmaceutically acceptable salt thereof, and instructions for using the compound or composition. In some embodiments, the kits comprise at least one disclosed compound and one or more cationic and/or ionizable lipids, phospholipids, neutral or non-cationic lipids, polyethylene glycol (PEG)-lipid conjugates, and/or structural lipids. The kit may further comprise at least one active agent. The kits can also comprise other agents and/or products co-packaged, co-formulated, and/or co-delivered with other components.

[0244] The kits can also comprise instructions for using the components of the kit. The instructions are relevant materials or methodologies pertaining to the kit. The materials may include any combination of the following: background information, list of components, brief or detailed protocols for using the compositions, trouble-shooting, references, technical support, and any other

related documents. Instructions can be supplied with the kit or as a separate member component, either as a paper form or an electronic form which may be supplied on computer readable memory device or downloaded from an internet website, or as recorded presentation.

[0245] It is understood that the disclosed kits can be employed in connection with the disclosed methods. The kit may further contain containers or devices for use with the methods or compositions disclosed herein. The kits optionally may provide additional components such as buffers and disposable single-use equipment (e.g., pipettes, cell culture plates or flasks).

[0246] The kits provided herein are in suitable packaging. Suitable packaging includes, but is not limited to, vials, bottles, jars, flexible packaging, and the like. Individual member components of the kits may be physically packaged together or separately.

9. Examples

[0247] Abbreviations used in the schemes and examples that follow are: BOC is tert-butyloxycarbonyl; DMAP is 4-dimethylaminopyridine; DMF is dimethylformamide; EDC is 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide; eq is equivalent; EtOAc is ethyl acetate; MeOH is methanol; RT or r.t. is room temperature; and THE is tetrahydrofuran.

[0248] All air and moisture sensitive manipulations were performed under either argon or in vacuo using standard Schlenk techniques. Anhydrous solvents (Et.sub.2O, THE, Dioxane, DMSO, DMF, DCM and Toluene) were purchased from Fischer Scientific. All chemicals were purchased from Fischer Scientific, Sigma Aldrich, TCI WUXI Apptec and DC Chemicals Europe and were used without further purification unless mentioned otherwise.

[0249] Analytical thin layer chromatography (TLC) was performed with Merck SIL G/UV254 plates. Compounds were visualized by exposure to UV light or by dipping the plates in solutions of ninhydrin or potassium permanganate followed by heating or by staining with Iodine vapor in a wide jar chamber. Column chromatography was performed in air with silicagel 60 (Fluka), Column chromatography was performed with Merck Kieselgel 60 (200-500 mm). The solvent systems were given (s/s v: v).

[0250] NMR spectra ¹H (300 MHz), ¹³C (75 MHz) were respectively recorded on an ARX 300 or an Avance II 500 Bruker spectrometer. Chemical shifts (δ, ppm) are given with reference to residual ¹H or ¹³C of deuterated solvents in the solvent indicated (CDCl.sub.3 7.26, 77.00; (CD.sub.3).sub.2CO 2.05, 29.84 and 206.26, (CD.sub.3).sub.2SO 2.50, 39.52)). ¹H- and ¹³C-NMR chemical shifts (δ) are quoted in parts per million (ppm) relative to the TMS scale. Coupling constants J are quoted in Hz. The following abbreviations are used for the proton spectra multiplicities: s: singlet, d: doublet, t: triplet, q: quartet, qt: quintuplet, m: multiplet, br.: broad, dd: double doublet, dt: double triplet. Coupling constants (J) are reported in Hertz (Hz). Several signals could not be attributed will be represented by ArH (aromatic hydrogen).

[0251] Mass spectra (MS) were recorded with an LCQ-advantage (ThermoFinnigan) mass spectrometer with positive (ESI+) or negative (ESI-) electrospray ionization (ionization tension 4.5 kV, injection temperature 240° C.).

Example 1

Compounds

[0252] General Procedure 1: Esterification. Under a nitrogen atmosphere, a solution of the carboxylic acid derivative (1 eq.), EDC hydrochloride (1 eq.), and DMAP (0.5 eq.) in dry THF (0.1 M) was stirred at 0° C. for 0.5 h. Then after, a solution of the alcohol (1 eq.) in dry THF (0.1 M) was added. The solution was stirred at 0° C. to r.t. for 16 h. Progress of the reaction was monitored by TLC using CH.sub.2—Cl.sub.2-MeOH 10:1 mixture as eluent and ¹H NMR using ARX 300 Bruker spectrometer. Upon completion of the reaction, the mixture was extracted with dichloromethane (3×). The combined organic layer and washed with 0.2 M HCl aqueous, saturated aqueous solution of NaHCO.sub.3 and brine, respectively. Then, the solid residue was purified by manual column chromatography using CH.sub.2Cl.sub.2-EtOAc or CH.sub.2Cl-MeOH step gradient solvent system as an eluent to afford the titled product.

(9Z,12Z)-octadeca-9,12-dien-1-yl 1-methyl-D-tryptophanate

##STR00023##

[0253] Step 1: (9Z,12Z)-octadeca-9,12-dien-1-yl(9Z,12Z)-octadeca-9,12-dien-1-yl Na-(tert-butoxycarbonyl)-1-methyl-D-tryptophanate was synthesized according to General Procedure 1 from N.sup.α-(tert-butoxycarbonyl)-1-methyl-D-tryptophan (1 eq), EDC. hydrochloride (1 eq.), DMAP (0.5 eq.) and linoleyl alcohol (1 eq) in dry THF (0.1 M) during 4 h at room temperature.

[0254] Step 2: (9Z,12Z)-octadeca-9,12-dien-1-yl 1-methyl-D-tryptophanate was synthesized from (9Z,12Z)-octadeca-9,12-dien-1-yl(9Z,12Z)-octadeca-9,12-dien-1-yl Na-(tert-butoxycarbonyl)-1-methyl-D-tryptophanate (1 eq) stirred in HCl solution of dioxane (HCl 4 M in dioxane) from 0° C. to RT during 5 h.

((4-((1-methyl-D-tryptophyl)oxy)butyl)azanediyl)bis(hexane-6,1-diyl)bis(2-hexyldecanoate)

##STR00024##

[0255] Step 1: ((4-((1-methyl-Na-pivaloyl-D-tryptophyl)oxy)butyl)azanediyl)bis(hexane-6,1-diyl)bis(2-hexyldecanoate) was synthesized according to General Procedure 1 from N.sup.α-(tert-butoxycarbonyl)-1-methyl-D-tryptophan (1 eq), EDC. hydrochloride (1 eq.), DMAP (0.5 eq.) and ALC-0315 (1 eq) in dry THF (0.1 M) during 4 h at room temperature.

[0256] Step 2: ((4-((1-methyl-D-tryptophyl)oxy)butyl)azanediyl)bis(hexane-6,1-diyl)bis(2-hexyldecanoate) was synthesized from ((4-((1-methyl-Na-pivaloyl-D-tryptophyl)oxy)butyl)azanediyl)bis(hexane-6,1-diyl)bis(2-hexyldecanoate) (1 eq) stirred in HCl solution of dioxane (HCl 4 M in dioxane) from 0° C. to RT during 5 h.

[0257] .sup.1H NMR (500 MHz, CDCl.sub.3) δ 7.59 (d, J=7.9 Hz, 1H), 7.28 (d, J=8.2 Hz, 1H), 7.21 (t, J=7.5 Hz, 1H), 7.09 (t, J=7.4 Hz, 1H), 6.92 (s, 1H), 4.16-4.00 (m, 8H), 3.74 (s, 3H), 3.26 (dd, J=14.4, 4.8 Hz, 1H), 3.01 (dd, J=14.4, 7.8 Hz, 1H), 2.49-2.34 (m, 7H), 2.29 (td, J=8.9, 4.5 Hz, 3H), 1.67-1.52 (m, 15H), 1.32-1.17 (m, 52H), 0.86 (t, J=6.8 Hz, 12H).

Heptadecan-9-yl 8-((4-((1-methyl-D-tryptophyl)oxy)butyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate

##STR00025##

[0258] Step 1: heptadecan-9-yl 8-((4-((1-methyl-Na-pivaloyl-D-tryptophyl)oxy)butyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate was synthesized according to General Procedure 1 from N.sup.α-(tert-butoxycarbonyl)-1-methyl-D-tryptophan (1 eq), EDC. hydrochloride (1 eq.), DMAP (0.5 eq.) and SM-102 (1 eq) in dry THF (0.1 M) during 4 h at room temperature.

[0259] Step 2: heptadecan-9-yl 8-((4-((1-methyl-D-tryptophyl)oxy)butyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate was synthesized from heptadecan-9-yl 8-((4-((1-methyl-Na-pivaloyl-D-tryptophyl)oxy)butyl)(6-oxo-6-(undecyloxy)hexyl)amino)octanoate (1 eq) stirred in HCl solution of dioxane (HCl 4 M in dioxane) from 0° C. to RT during 5 h.

[0260] .sup.1H NMR (500 MHz, CDCl.sub.3) δ 7.58 (d, J=7.9 Hz, 1H), 7.27 (d, J=7.7 Hz, 1H), 7.20 (t, J=7.5 Hz, 1H), 7.09 (t, J=7.4 Hz, 1H), 6.92 (s, 1H), 4.89-4.80 (m, 1H), 4.15-4.08 (m, 2H), 4.06-4.01 (m, 2H), 3.80-3.75 (m, 1H), 3.74 (s, 3H), 3.25 (dd, J=14.4, 4.8 Hz, 1H), 3.02 (dd, J=14.4, 7.6 Hz, 1H), 2.62 (t, J=6.4 Hz, 2H), 2.40 (dt, J=8.0, 5.3 Hz, 4H), 2.26 (dt, J=10.6, 7.5 Hz, 5H), 1.65-1.55 (m, 7H), 1.54-1.45 (m, 5H), 1.44-1.35 (m, 5H), 1.36-1.17 (m, 49H), 0.87 (t, J=7.0 Hz, 9H).

Example 2

Lipid Compositions

[0261] Exemplary lipid compositions include any or all of: nucleic acid (such as mRNA, aptamers, antisense oligonucleotides, ribozyme nucleic acids, RNA interference and antigene nucleic acids); modified lipids as disclosed herein; phospholipids (for example, phosphatidylcholine and phosphatidylethanolamine); cholesterol; and polyethylene glycol (PEG)-functionalized lipids (PEG-lipids).

[0262] Cationic liposomes are generated with the dry film method. Lipids and cholesterol are dissolved in chloroform in a round-bottom flask. After removing the organic solvent by rotary

evaporation at 40° C. the dry lipid firm is then hydrated in PBS (pH 7.4) with six cycles of vortexing for 30 s every 5 min at 45° C. The resulting lipid suspension is freeze-thawed 10 cycles between liquid nitrogen and 37° C. water bath and extruded by an Extruder with 0.2 μm polycarbonate filter membranes to obtain the homogeneous nanosuspension. Nucleic acids are then mixed with the liposomes and incubated for 20 min at RT to allow sufficient encapsulation. [0263] The nucleic acids are prepared in acetate buffer at pH 4.0. Depending on the desired formulation, an ethanol solution containing lipids at the appropriate molar ratio solutions are prepared.

[0264] The diameter, size distribution, polydispersity index (PDI), and zeta-potential are determined with a ZETASIZER Nano (ZEN3600, Malvern Instruments, Malvern, WR, UK) and the JEOL2010F transmission electron microscope (TEM) and the FEI Quanta 200FEG scanning electron microscope (Phillips FEI Co., The Netherlands) are used to visualize the morphology of the lipid nanoparticles and liposomes.

[0265] To detect the encapsulation efficiency (EE) and drug loading (DL), 40 μL of the final composition is withdrawn and mixed with 160 μL acetonitrile, followed by vortexing for 15 s and water sonication for 5 min. After centrifugation for 8 min at 12,000 rpm, the supernatant is collected for high performance liquid chromatography (HPLC) measurement. The EE and DL is calculated by the following equations:

$$[00001] EE(\%) = \frac{ma - mb}{ma} \times 100\% \quad DL(\%) = \frac{ma - mb}{(ma + 100mg)} \times 100\%$$

where ma indicates the amount of active agent (e.g., nucleic acids) in the initial chloroform solution, mb represents the content of active agent in the lipid nanoparticles and liposomes detected by the HPLC.

Example 3

In Vitro and In Vivo Transfection, Pharmacokinetics, and Efficacy

[0266] In vitro and in vivo transfection can be detected in cells or in mice. To study the pharmacokinetics of the formulations, wide type C57BL/6 mice (male, 8 weeks) are randomly assigned to different groups and the formulations are administered at 5 min, 0.5 h, 4 h, 8 h, 24 h, and 48 h post administration, blood is collected and plasma is isolated after centrifugation (13,300 rpm, 10 min). Subsequently, the mice are euthanized and major organs including the liver, spleen, lung, lymph node, fat pad, and tumor are harvested, weighed, and homogenized. The content of active agent in the tissue homogenate and plasma are detected by the liquid chromatography-mass spectrometry (LC-MS) using an ABI-5500 Qtrap (Sciex, Ontario, Canada) mass spectrometer with electrospray ionization source interfaced with Shimadzu high performance liquid chromatography (HPLC) system with Xbridge C18 column (50×2.1 mm ID, 3.5 μm). The efficacy of formulations is evaluated on different disease models.

[0267] It is understood that the foregoing detailed description and accompanying examples are merely illustrative and are not to be taken as limitations upon the scope of the disclosure, which is defined solely by the appended claims and their equivalents.

[0268] Various changes and modifications to the disclosed embodiments will be apparent to those skilled in the art and may be made without departing from the spirit and scope thereof.

Claims

1. A compound of formula (I) ##STR00026## or a pharmaceutically acceptable salt thereof, wherein: R.sup.1 is hydrogen, or C.sub.1-C.sub.6 alkyl; X is selected from O, NR.sup.2, S, and a bond; R.sup.2 is hydrogen C.sub.1-C.sub.6 alkyl, C.sub.2-C.sub.6 alkenyl, C.sub.2-C.sub.6 alkynyl, C.sub.3-C.sub.6 cycloalkyl, or halo-C.sub.1-C.sub.6-alkyl; and L is selected from C.sub.8-C.sub.80 alkyl, C.sub.8-C.sub.80 alkenyl, C.sub.8-C.sub.80 alkynyl, C.sub.8-C.sub.80 heteroalkyl, C.sub.8-C.sub.80 heteroalkenyl, and C.sub.8-C.sub.80 heteroalkynyl, each of which is optionally substituted with one or more substituents selected from hydroxy and amino.

2. The compound of claim 1, or a pharmaceutically acceptable salt thereof, wherein R^{sup.1} is methyl.
3. The compound of claim 1 or 2, or a pharmaceutically acceptable salt thereof, wherein L is selected from C.sub.12-C.sub.40 alkyl and C.sub.12-C.sub.40 alkenyl.
4. The compound of any of claims 1-3, or a pharmaceutically acceptable salt thereof, wherein L has a formula (A): ##STR00027## wherein: n is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, or 40; and R^{sup.a} and R^{sup.b} are each independently selected from C.sub.6-C.sub.40 alkyl, C.sub.6-C.sub.40 alkenyl, C.sub.6-C.sub.40 heteroalkyl, and C.sub.6-C.sub.40 heteroalkenyl.
5. The compound of claim 4, or a pharmaceutically acceptable salt thereof, wherein n is 1, 2, 3, 4, 5, 6, 7, or 8.
6. The compound of any one of claims 1-5, or a pharmaceutically acceptable salt thereof, wherein L has a formula (D), (E), or (F): ##STR00028## wherein: n, p, and q are each independently 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, or 40; and R^{sup.a1} and R^{sup.a2} are each independently selected from C.sub.6-C.sub.40 alkyl and C.sub.6-C.sub.40 alkenyl.
7. The compound of claim 6, or a pharmaceutically acceptable salt thereof, wherein n, p, and q are each independently 1, 2, 3, 4, 5, 6, 7, or 8.
8. The compound of any one of claims 1-7, or a pharmaceutically acceptable salt thereof, wherein L is selected from: ##STR00029## ##STR00030## ##STR00031##
9. The compound of claim 1, wherein the compound is selected from the group consisting of: ##STR00032##
10. A lipid composition comprising a compound of any of claims 1-9.
11. The lipid composition of claim 10, further comprising one or more of a cationic and/or ionizable lipid, a phospholipid, a neutral or non-cationic lipid, or a combination thereof.
12. The lipid composition of claim 10 or 11, further comprising a structural lipid.
13. The lipid composition of claim 12, wherein the structural lipid is selected from the group consisting of cholesterol, fecosterol, sitosterol, ergosterol, campesterol, stigmasterol, brassicasterol, tomatidine, cholesteryl hemisuccinate, cholesteryl sulfate, ursolic acid, alpha-tocopherol, and mixtures thereof.
14. The lipid composition of any of claims 10-13, further comprising a polyethylene glycol (PEG)-lipid conjugate.
15. The lipid composition of any of claims 10-14, further comprising at least one active agent.
16. The lipid composition of claim 15, wherein the at least one active agent is encapsulated by a liposome, a lipid nanoparticle, a micelle in the lipid composition.
17. The lipid composition of claim 15 or 16, wherein the at least one active agent comprises a nucleic acid or protein.
18. The lipid composition of any of claims 15-17, wherein the at least one active agent comprises an RNA.
19. The lipid composition of claim 18, wherein the RNA is selected from the group consisting of a small interfering RNA (siRNA), an asymmetrical interfering RNA (aiRNA), a microRNA (miRNA), a Dicer-substrate RNA (dsRNA), a small hairpin RNA (shRNA), a messenger RNA (mRNA), a ribozyme, and mixtures thereof.
20. The lipid composition of any of claims 17-19, wherein the nucleic acid encodes an antigen or a functional fragment thereof.
21. The lipid composition of any of claims 10-20, wherein the at least one active agent comprises an immune modulator, a chemotherapeutic agent, a steroid, an analgesic, an antimicrobial agent, or a combination thereof.
22. A pharmaceutical composition comprising the lipid composition of any of claims 10-21 and a pharmaceutically acceptable carrier.

- 23.** A vaccine comprising the lipid composition of any of claims 10-21 and an antigen or a nucleic acid encoding thereof.
- 24.** The vaccine of claim 23, wherein the antigen is a tumor antigen, a self-antigen, or an infectious disease derived antigen.
- 25.** The vaccine of claim 23 or 24, wherein the nucleic acid is messenger RNA (mRNA).
- 26.** A method for delivering an active agent to a cell comprising contacting the cell with the lipid composition of any of claims 10-21 or the pharmaceutical composition of claim 22.
- 27.** A method for delivering an active agent to a subject comprising administering to the subject the lipid composition of any of claims 10-21 or the pharmaceutical composition of claim 22.
- 28.** A method of producing a polypeptide of interest in a cell, the method comprising contacting the cell with the lipid composition of any of claims 10-21 or the pharmaceutical composition of claim 22, wherein the lipid composition comprises an mRNA encoding the polypeptide of interest.
- 29.** The method of claim 28, wherein the cell is in a subject and the contacting comprises administering to the subject.
- 30.** A method of treating or preventing a disease or disorder in a subject comprising administering to the subject in need thereof an effective amount of the lipid composition of any of claims 10-21, the pharmaceutical composition of claim 22, or the vaccine of any of claims 23-25.
- 31.** The method of claim 30, wherein the disease or disorder comprises cancer, an autoimmune disease, an inflammatory disease, or an infectious disease.
- 32.** The method of claim 30 or 31, wherein the disease or disorder is cancer.
- 33.** The method of claim 31 or 32, wherein the subject has cancer, has had cancer, is predisposed to cancer, or has a family history of cancer.
- 34.** The method of any of claims 31-33, wherein the cancer comprises a solid tumor or hematological cancer.
- 35.** The method of any of claims 31-34, wherein the cancer is metastatic cancer.
- 36.** The method of any of claims 31-35, wherein the method suppresses or eliminates cancer metastasis, decreases tumor growth, prevents tumor recurrences, or any combination thereof.
- 37.** The method of any of claims 30-36, wherein the administering comprises an initial administration and at least one subsequent administration.
- 38.** The method of any of claims 30-37, wherein the subject is human.
- 39.** The method of any of claims 30-38, further comprising administering at least one additional active agent.
- 39.** The method of claim 38, wherein the at least one additional active agent comprises an immune modulator, a chemotherapeutic agent, a nucleic acid, a steroid, an analgesic, an antimicrobial agent, or a combination thereof.
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